

TECHNOLOGY



Designing Microsoft Azure Infrastructure Solutions: AZ:305

Design for High Availability



Learning Objectives

By the end of this lesson, you will be able to:

- Examine how availability sets are used to logically group VMs within a data center for improved reliability
- List the advantages of Azure Front Door in enhancing the speed, security, and scalability of web applications
- Analyze the various routing methods offered by Azure Traffic Manager and their effects on web app traffic distribution
- Review the use of Azure virtual machine scale sets for deploying identical VMs efficiently



Learning Objectives

By the end of this lesson, you will be able to:

- Examine the key components of the accelerated database recovery process and its impact on database availability
- Identify the different NoSQL database models and their applications in data management
- Evaluate the process of storage account failover and its role in mitigating service disruptions



Identify the Availability Requirements for Azure Resources

Delivering Reliability in Azure

The key to improving the reliability of a solution is to design it to continue to function despite failure.

In cloud solutions, survival through failure is a shared responsibility with responsibilities categorized into three levels, such as:

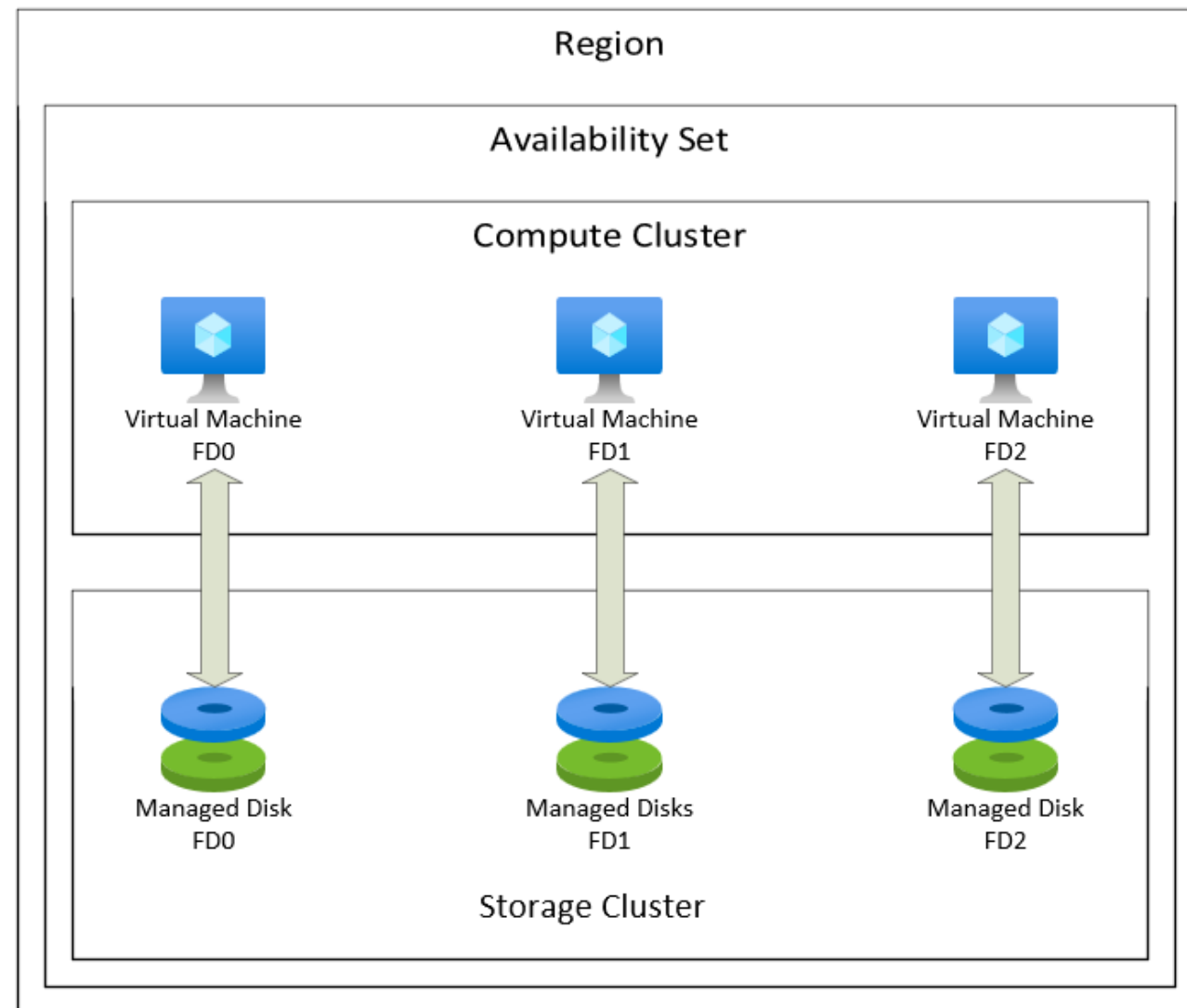
Resilient foundation

Resilient services

Resilient applications

Availability Sets

It is a logical grouping of VMs within a data center.

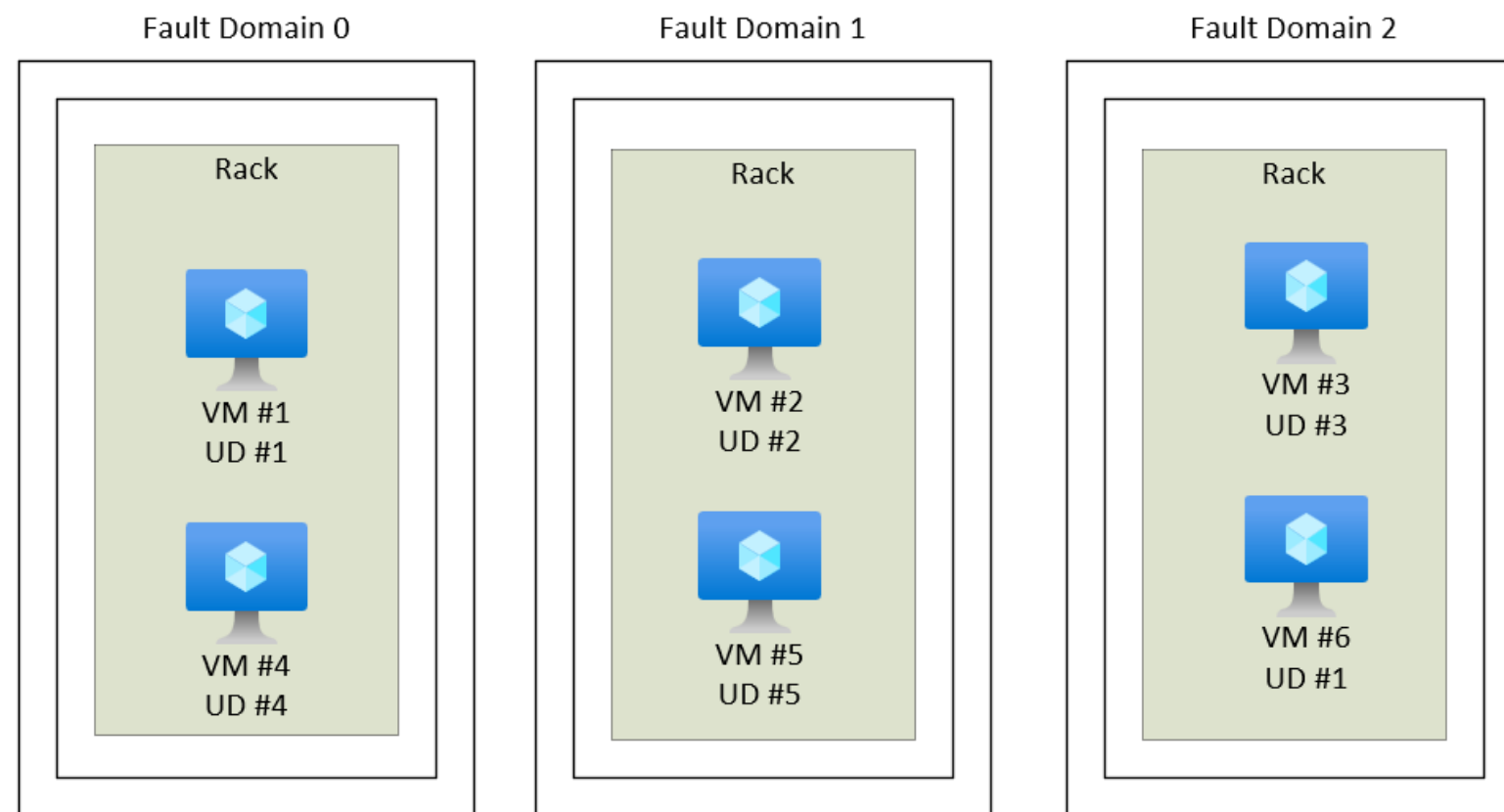


- On compute clusters, VMs are automatically distributed across fault and update domains.
- On storage clusters, it provides equal resiliency with managed disks.

Source: <https://learn.microsoft.com/en-us/azure/virtual-machines/availability-set-overview>

How Do Availability Sets Work?

Each virtual machine in an availability set is assigned an **update domain** and a **fault domain** by the underlying Azure platform.



Each availability set can be configured with up to 3 fault domains and 20 update domains.

How Do Availability Sets Work?

Update domains

- Azure can make incremental or rolling updates across a deployment using update domains.
- Only one update domain is rebooted at a time during scheduled maintenance.
- Five update domains are available by default, and up to 20 more can be added.

Fault domains

- These are a set of virtual machines that share a single point of failure and a similar set of hardware and switches.
- At least two fault domains are allocated to each VM in an availability set.
- Up to three fault domains can be configured.



Service Level Agreements (SLA)

It describes Microsoft's commitments to uptime and connectivity for Microsoft Online Services.
The percentage of SLAs in Azure are:

SLA 01

99.99%
Uptime guarantee

For all virtual machines with two or more instances deployed in two or more Availability Zones

SLA 02

99.95%
Uptime guarantee

For all virtual machines with two or more instances deployed in the same Availability Set

SLA 03

99.90%
Uptime guarantee

For any single instance virtual machine that uses premium storage for all operating system disks and data disks

High Availability and Azure SQL Database

The two types of high availability models used in the Azure SQL database are:



Standard
availability model

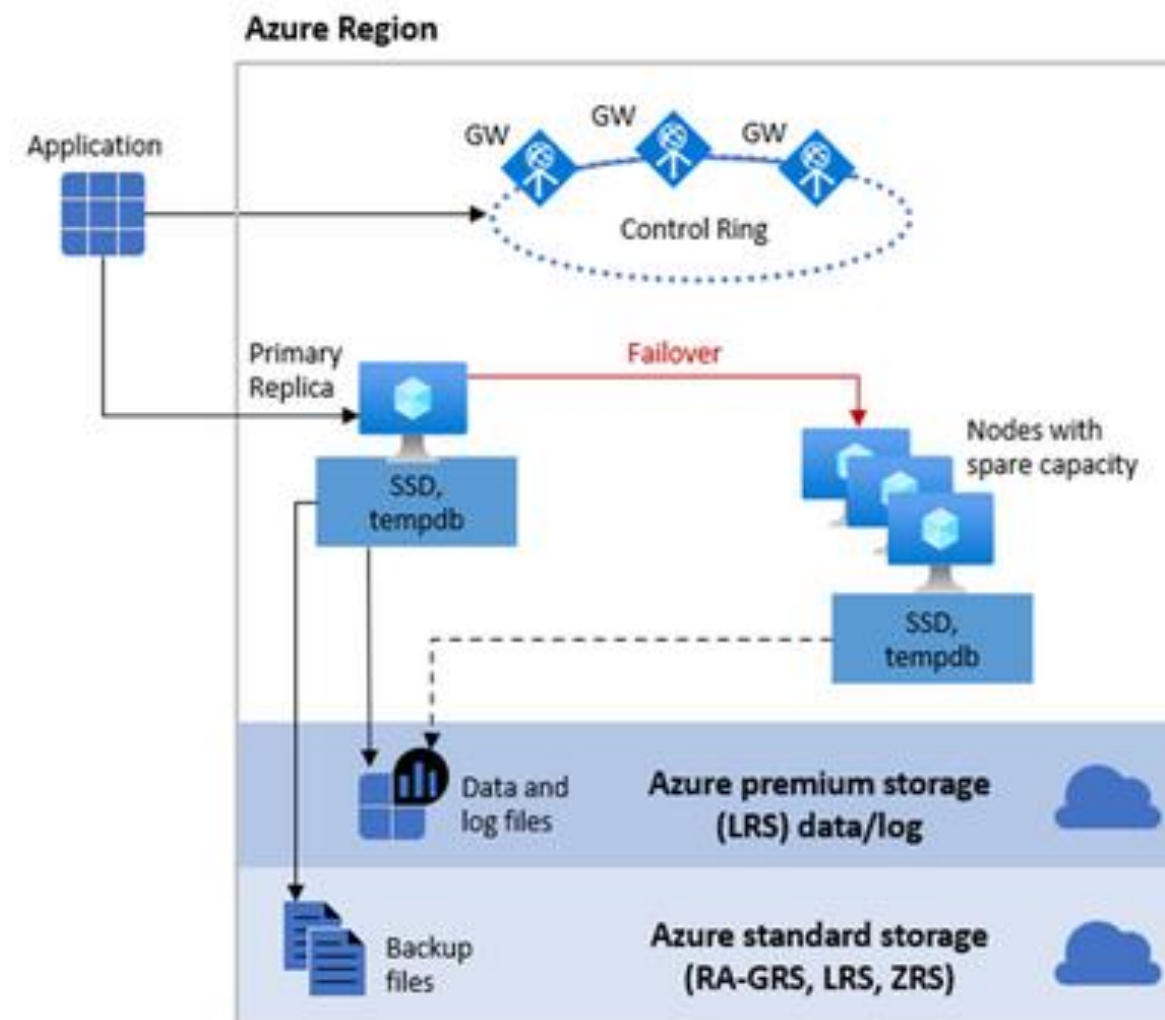


Premium
availability model

The standard availability model is based on a separation of compute and storage, whereas the premium availability model is based on a cluster of database engine processes.

Standard Availability Model

The following figure shows four nodes with separate compute and storage layers:



Source: <https://techcommunity.microsoft.com/t5/azure-sql-blog/high-availability-in-azure-sql-mi-general-purpose-service-tier/ba-p/3298977>

Standard Availability Model

It includes two layers:

Stateless compute
layer

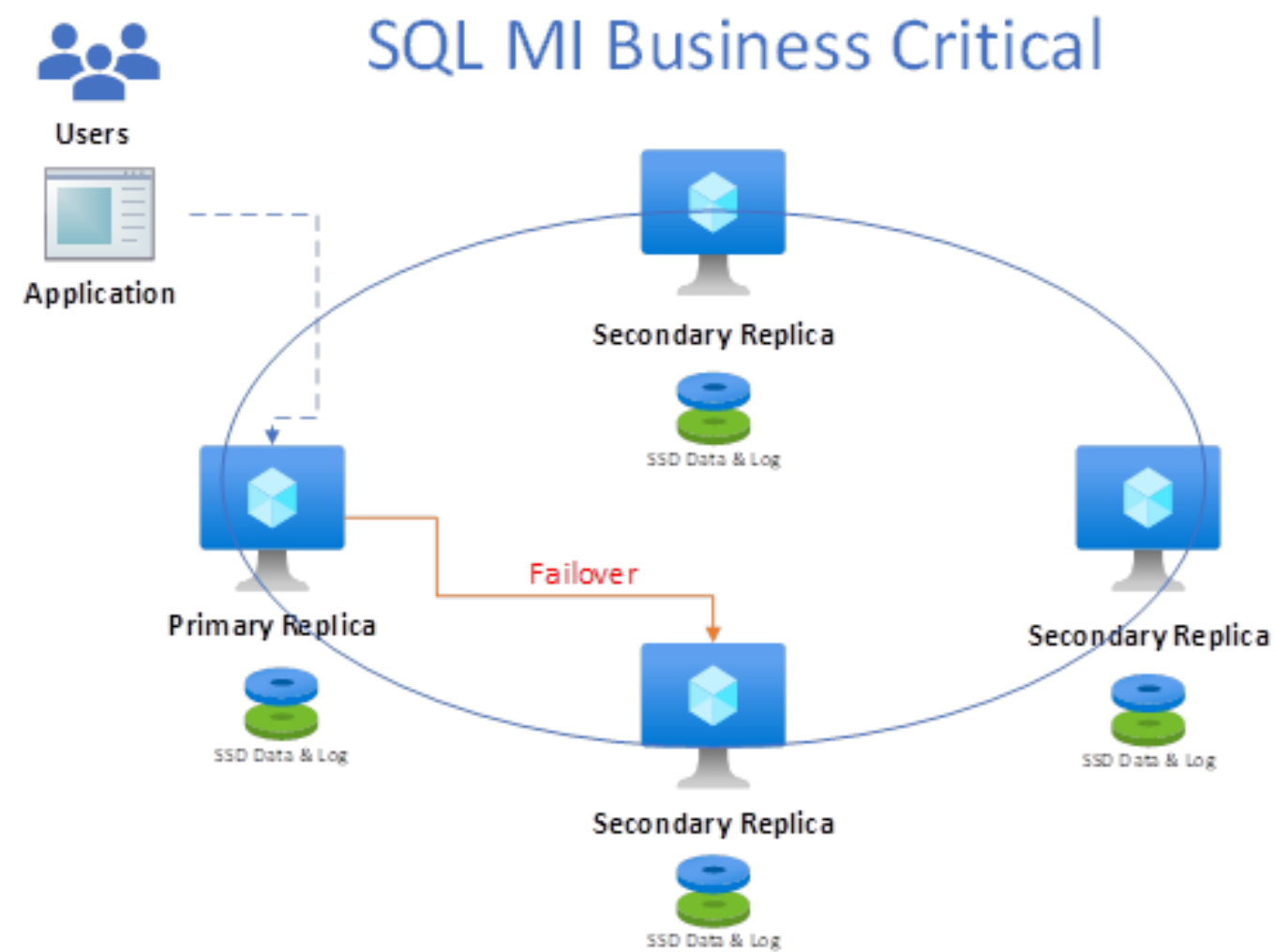
Runs the sqlservr.exe process and contains only transient and cached data

Stateful data layer

Guarantees that every record in the log file or page in the data file will be preserved even if the sqlservr.exe process crashes

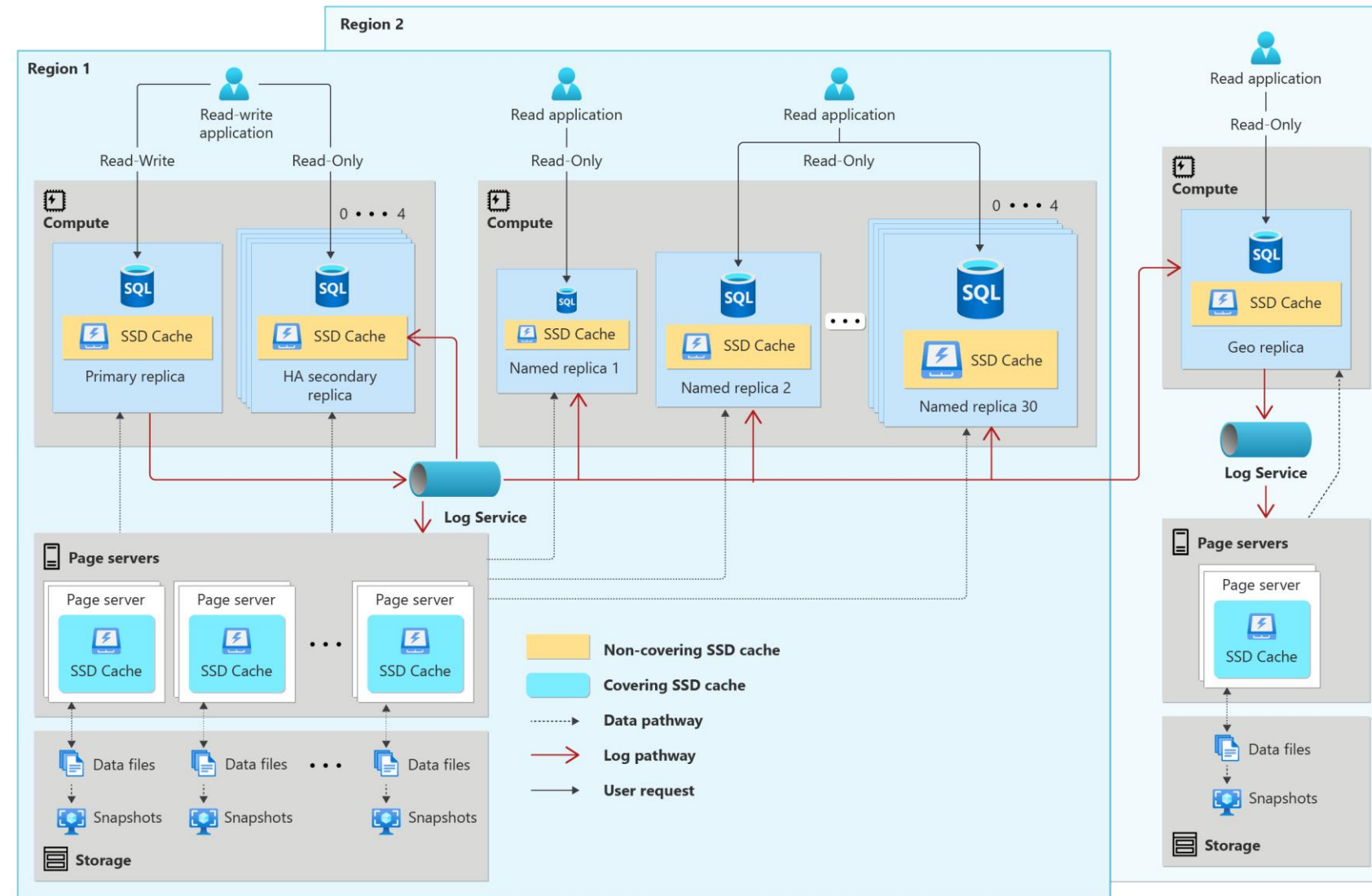
High Availability in the Business Critical Service Tier

The following diagram shows the architecture of the High Availability solution:



Hyperscale Service Tier Availability

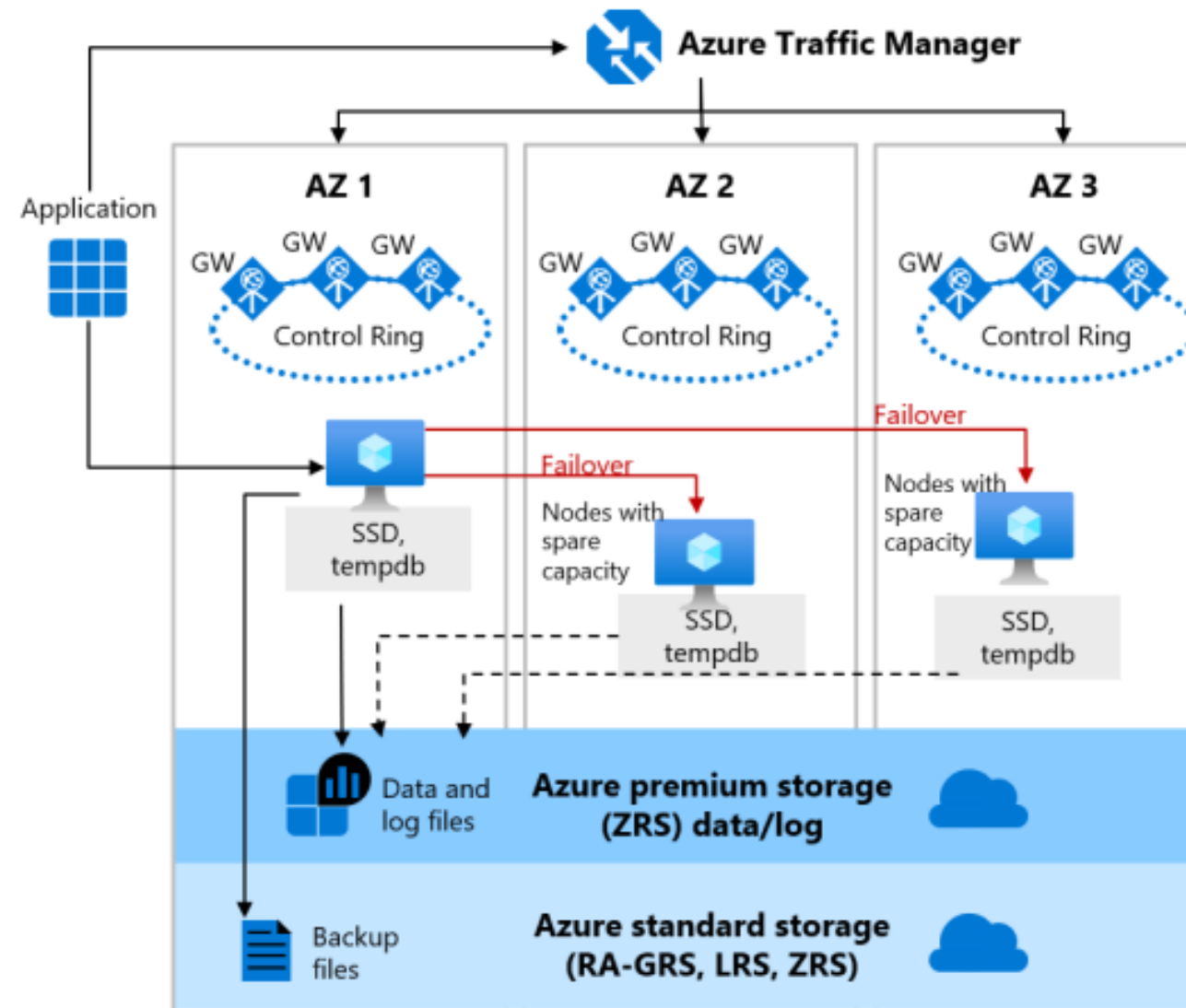
The following diagram illustrates the functional Hyperscale architecture:



Source: <https://learn.microsoft.com/en-us/azure/azure-sql/database/service-tier-hyperscale?view=azuresql>

Zone Redundant Configuration

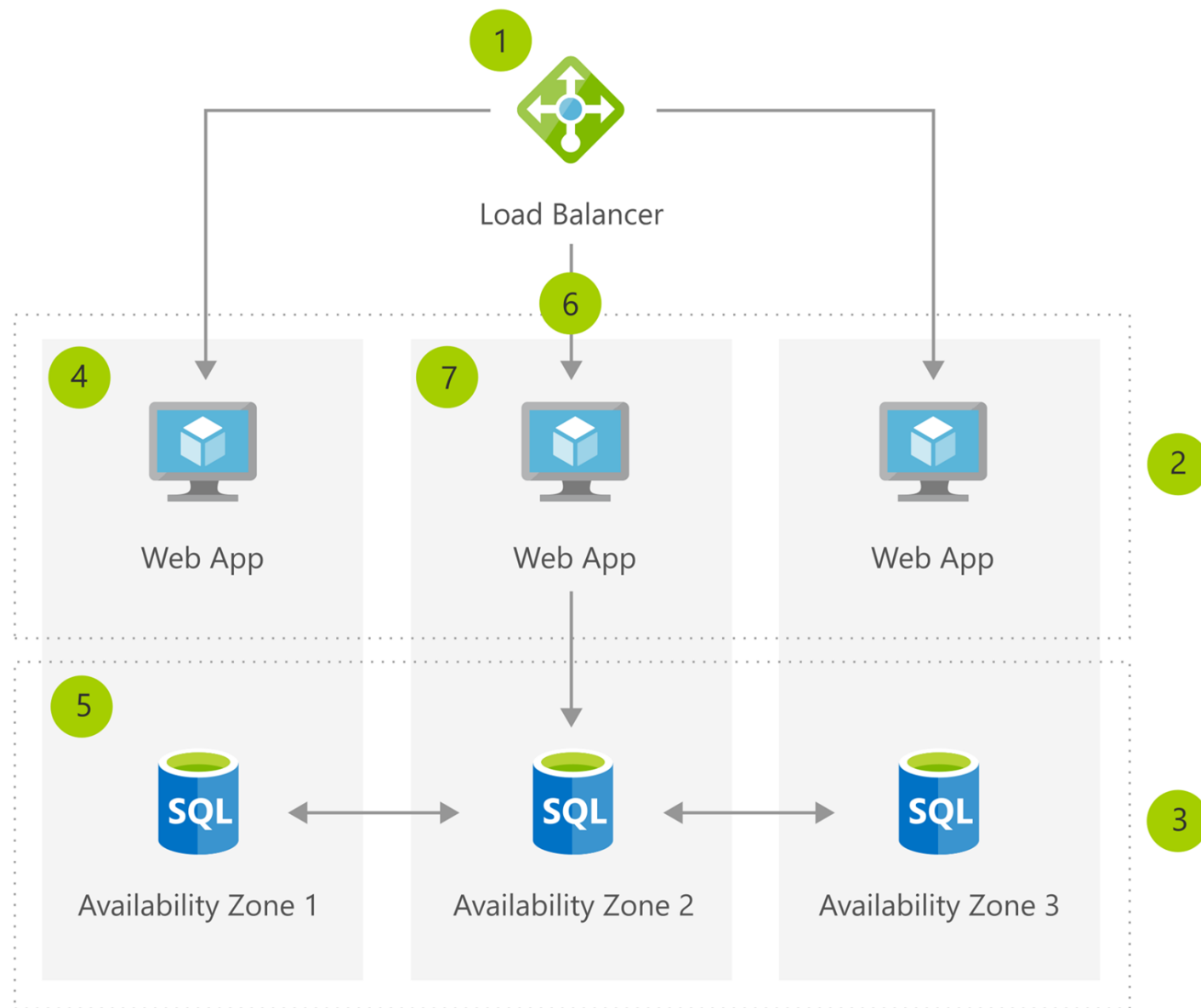
The following diagram illustrates the zone redundant version of high availability:



Source: <https://learn.microsoft.com/en-us/azure/azure-sql/database/high-availability-sla?view=azuresql&tabs=azure-powershell>

High Availability for Business Continuity and Disaster Recovery

This can be achieved by:



- Creating a zone-redundant load balancer
- Creating a front-end subnet
- Creating a DB subnet
- Creating VMs in three availability zones
- Configuring zone-redundant SQL DB
- Adding VMs to the load balancer's backend pool
- Deploying the application on VMs for redundancy and high availability

Configuring Virtual Machine Availability Set



Duration: 10 Min.

Problem Statement:

You have been requested to offer a method for achieving high availability for a virtual machine-hosted application. To do so, create an availability set, which is a logical grouping of VMs that helps Azure to understand how the application is designed to offer redundancy and availability.

Outcome:

You will be able to successfully demonstrate the configuration of an Azure VM Availability Set to ensure optimal redundancy and availability for a critical application.

Note: Refer to the demo document for detailed steps:
[01_Configuring_Virtual_Machine_Availability_Set](#)

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed are:

1. Log in to the Azure portal
2. Search for virtual machines
3. Configure availability set while creating a VM



Implementing Virtual Machine Availability Zone



Duration: 10 Min.

Problem Statement:

You have been asked to showcase the availability zone option available on your virtual machine. As an Azure Architect, this enhances the level of control over the availability of applications and data on your VMs.

Outcome:

You will be able to effectively demonstrate how leveraging availability zones enhances control and resilience of application and data availability across Azure VMs.

Note: Refer to the demo document for detailed steps:
[02_Implementing_Virtual_Machine_Availability_Zone](#)

UNASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed are:

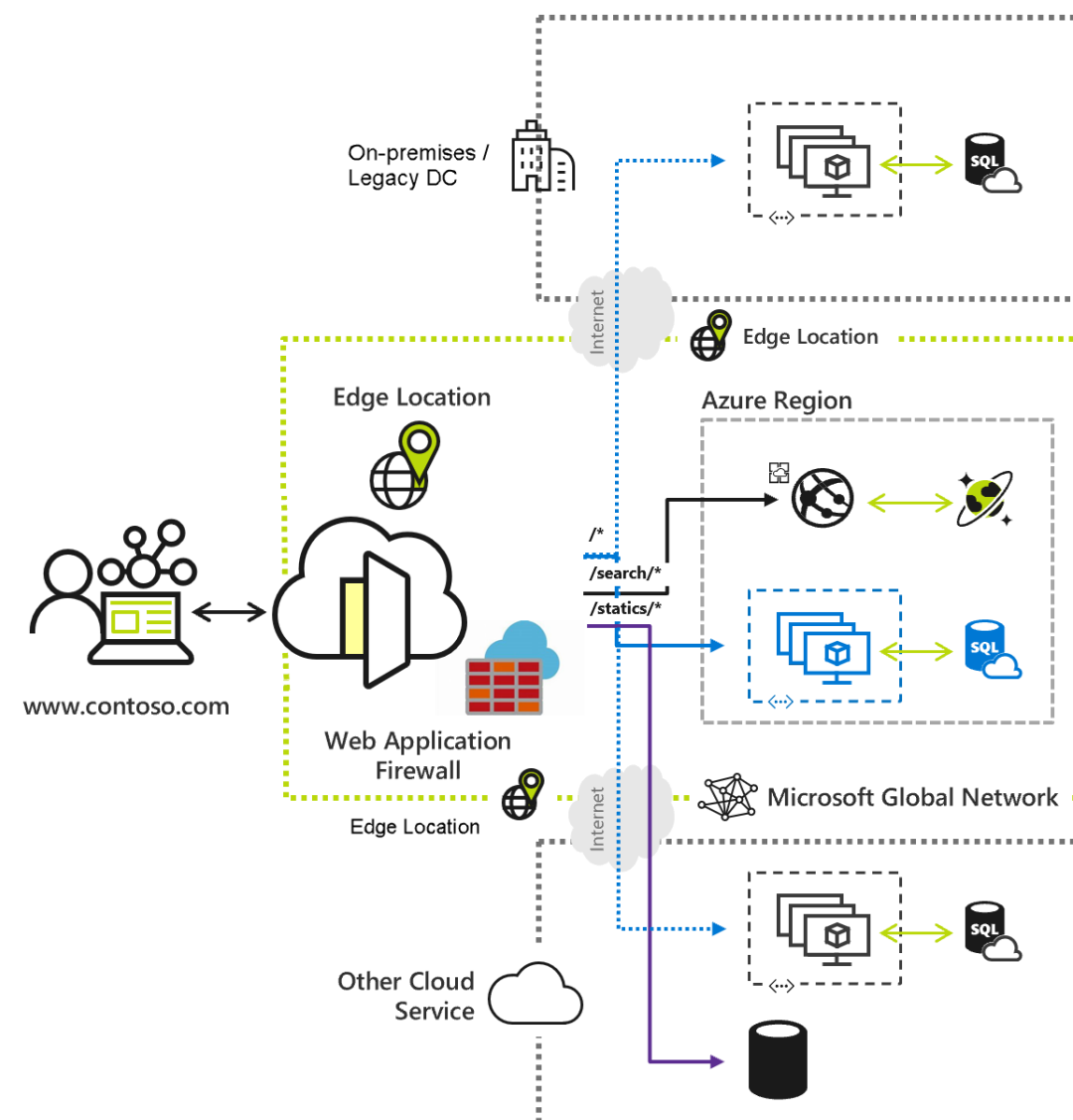
1. Log in to the Azure portal
2. Search and select virtual machines
3. Select availability zone
4. Configure the availability zone while creating a VM



Azure Front Door

Azure Front Door

It is a global, scalable entry-point that uses the Microsoft global edge network to create fast, secure, and highly scalable web applications.



Source: <https://learn.microsoft.com/en-us/azure/frontdoor/front-door-overview>

Azure Front Door: Benefits

Ensures efficient content delivery worldwide

Provides automatic failover and regional failover capabilities to minimize downtime

Integrates with the Azure Web Application Firewall for robust protection

Utilizes intelligent routing based on health, geography, and policies for optimized load balancing

Offers real-time insights into web traffic for performance monitoring and data-driven decisions



Azure Front Door: Features

It offers different capabilities across its Standard, Premium, and Classic tiers. For detailed information available at each tier, you can refer to the official [documentation here](#).

Global load
balancing

Traffic routing
policies

High
availability

Real-time
analytics

Custom
domains

Dynamic site
acceleration

Scalable
architecture



Creating Azure Front Door Service



Duration: 10 Min.

Problem Statement:

You have been requested to suggest a global, scalable entry-point that uses the Microsoft global edge network to create fast, secure, and widely scalable web applications.

Outcome:

You will be able to efficiently demonstrate Azure Front Door as a global entry-point, leveraging Microsoft's extensive edge network to ensure fast, secure, and scalable delivery of web applications.

Note: Refer to the demo document for detailed steps:
[03_Creating_Azure_Front_Door_Service](#)

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed are:

1. Configure Azure Front Door
2. Configure Frontend and Backend
3. Add a routing rule



Azure Traffic Manager

Azure Traffic Manager

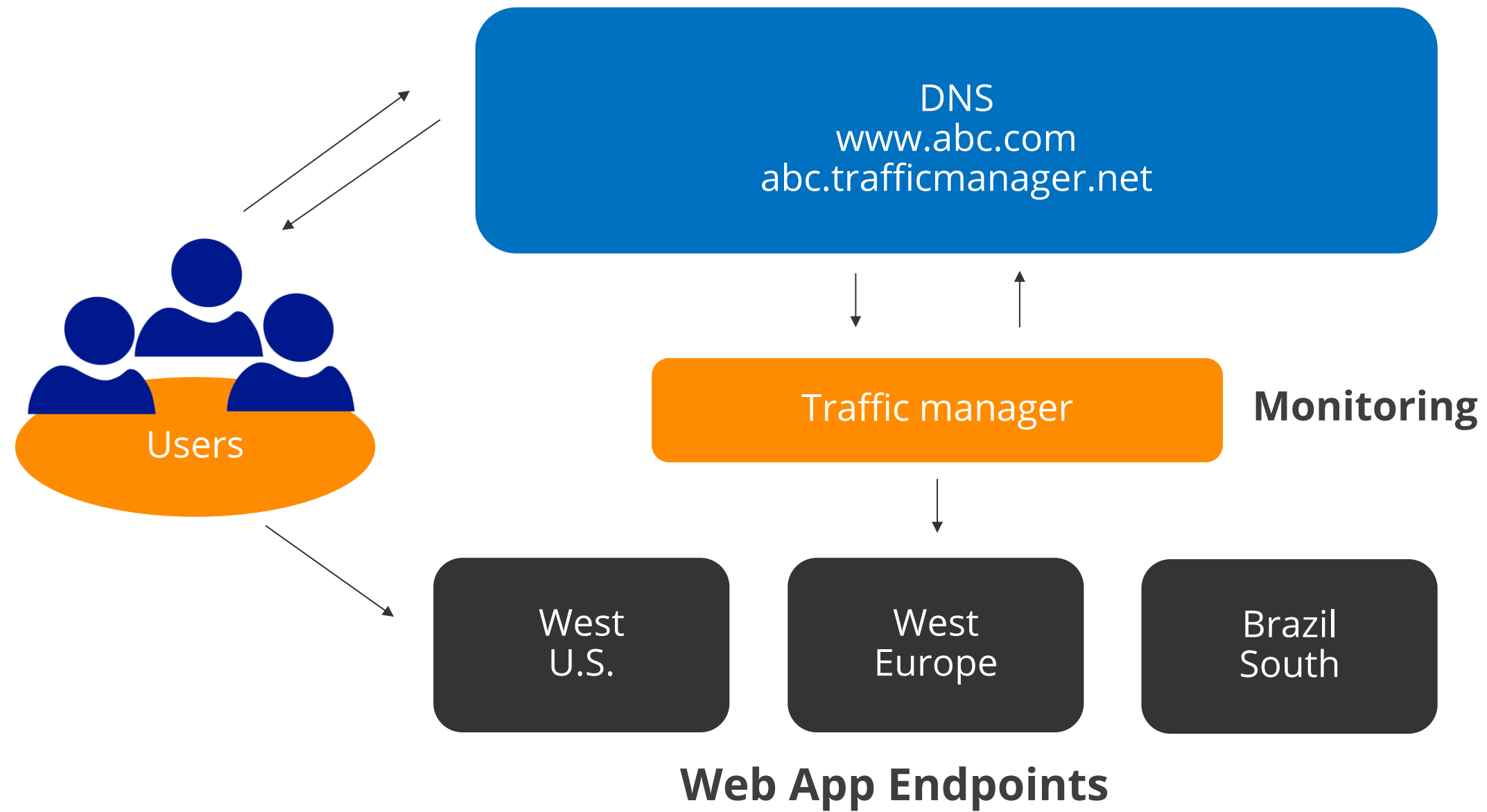
It is a network service that routes users to web app endpoints (deployments) in potentially different data centers located around the world.

Features

- Increases application availability
- Improves application performance
- Implements service maintenance without downtime
- Combines hybrid applications
- Distributes traffic for complex deployments



Traffic Manager Architecture



Traffic Manager Routing Methods

The following traffic routing methods are available in Traffic Manager:

Priority

Performance

Weighted

Geographic

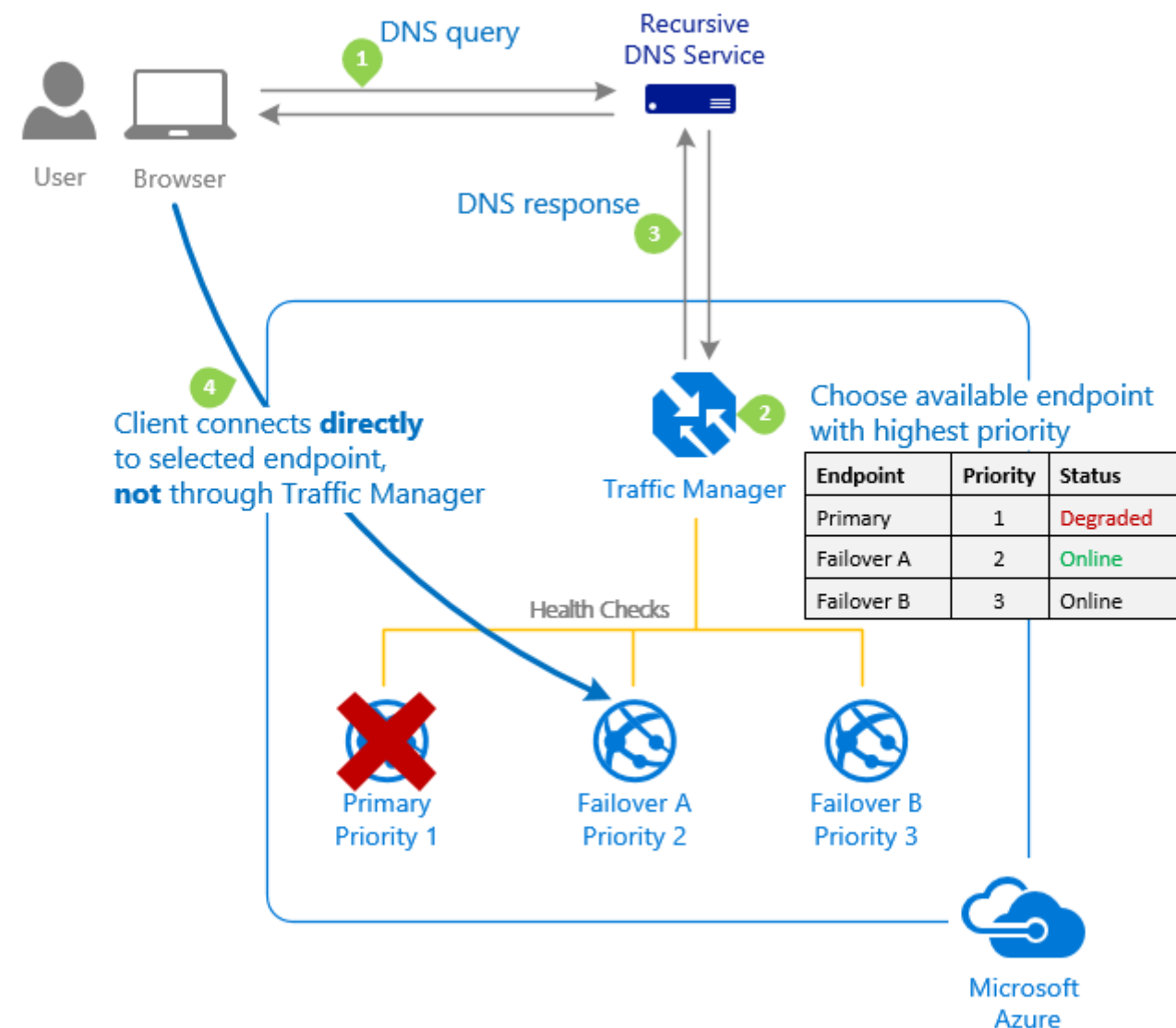
Subnet

Multivalue



Priority Traffic-Routing

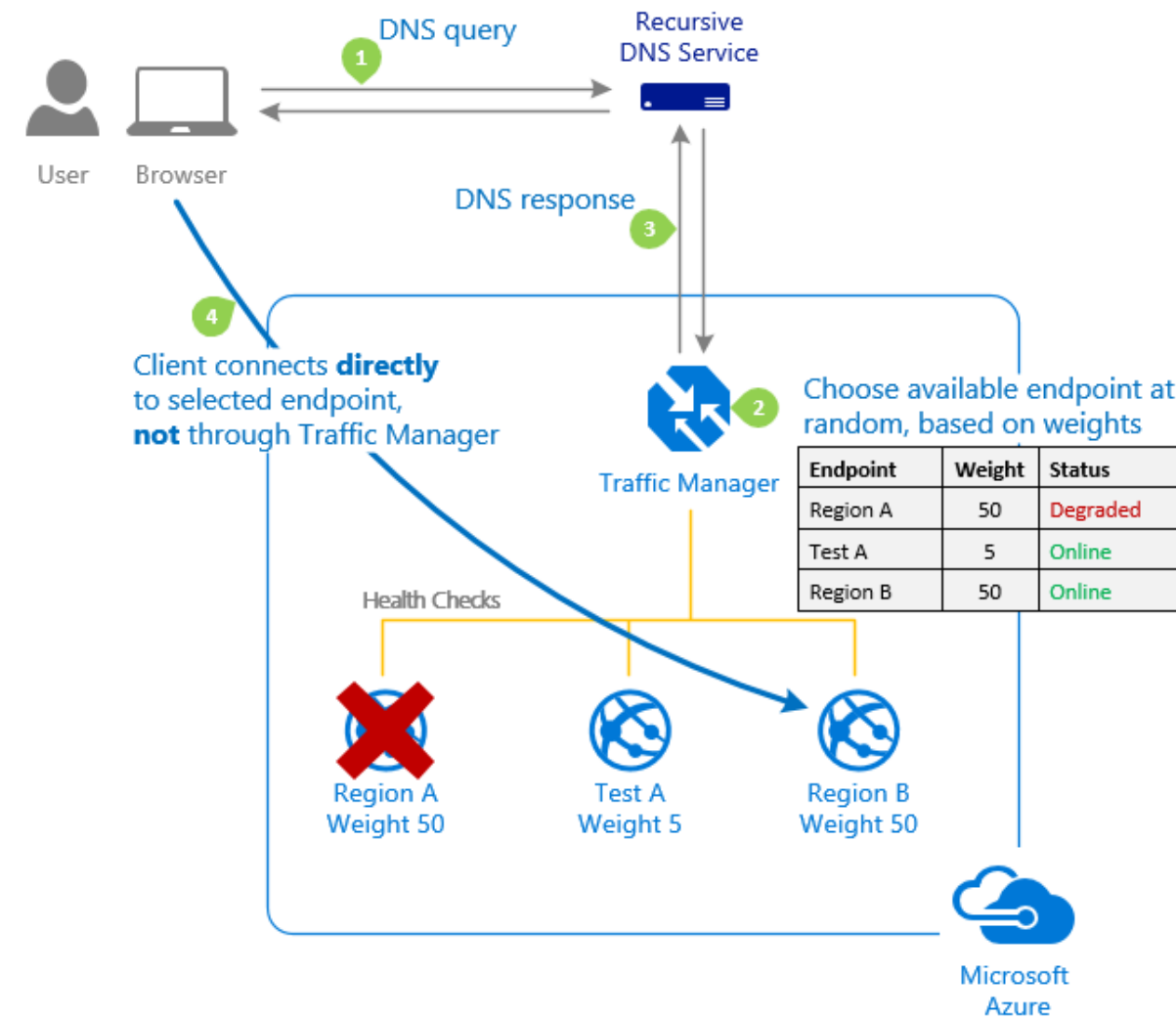
This method allows organizations to ensure service reliability by deploying backup services and easily implementing failover patterns for improved service continuity.



Source: <https://learn.microsoft.com/en-us/azure/traffic-manager/traffic-manager-routing-methods>

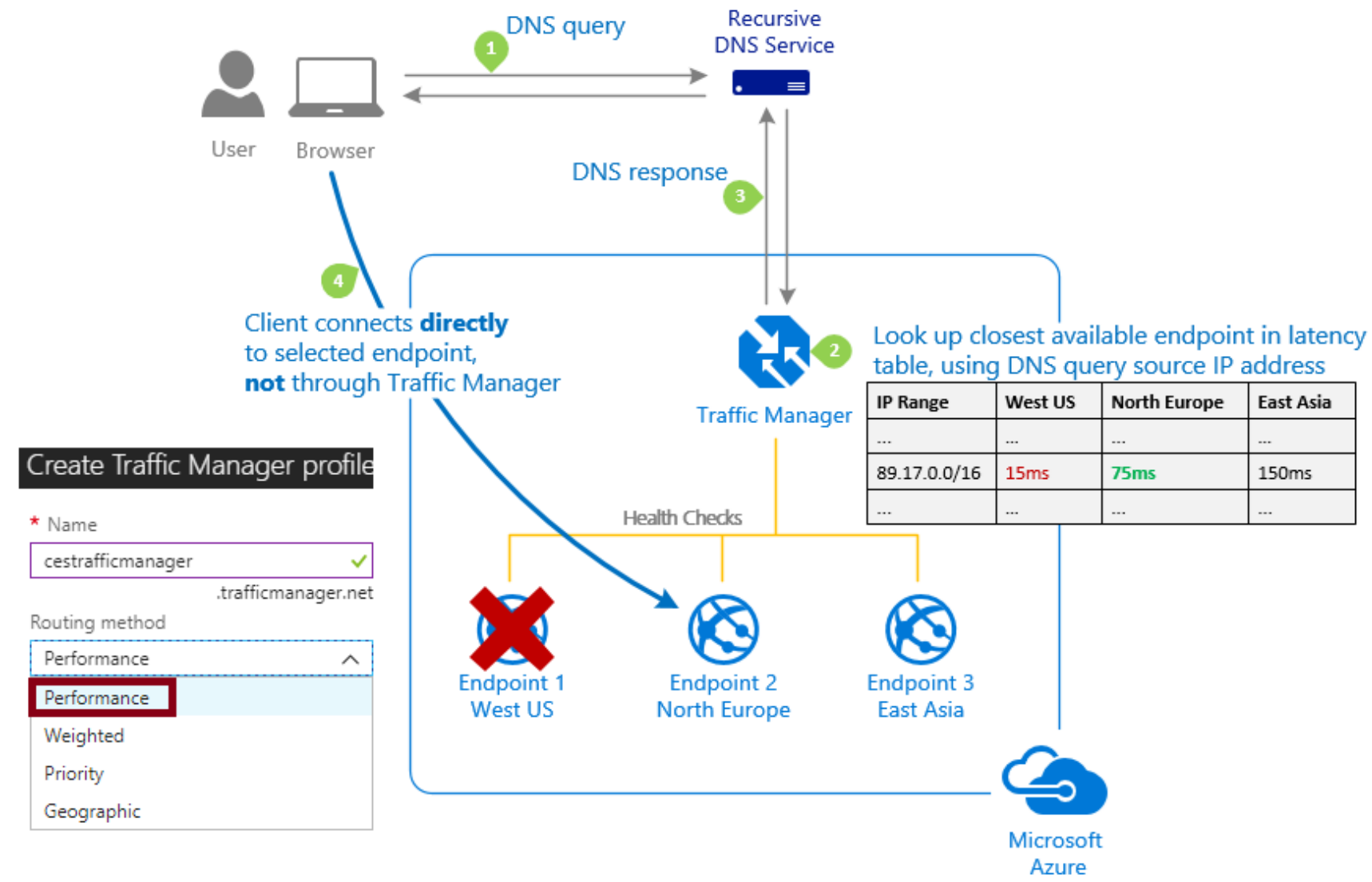
Weighted Traffic-Routing

This method distributes traffic evenly or according to a pre-defined weighting.



Performance Routing

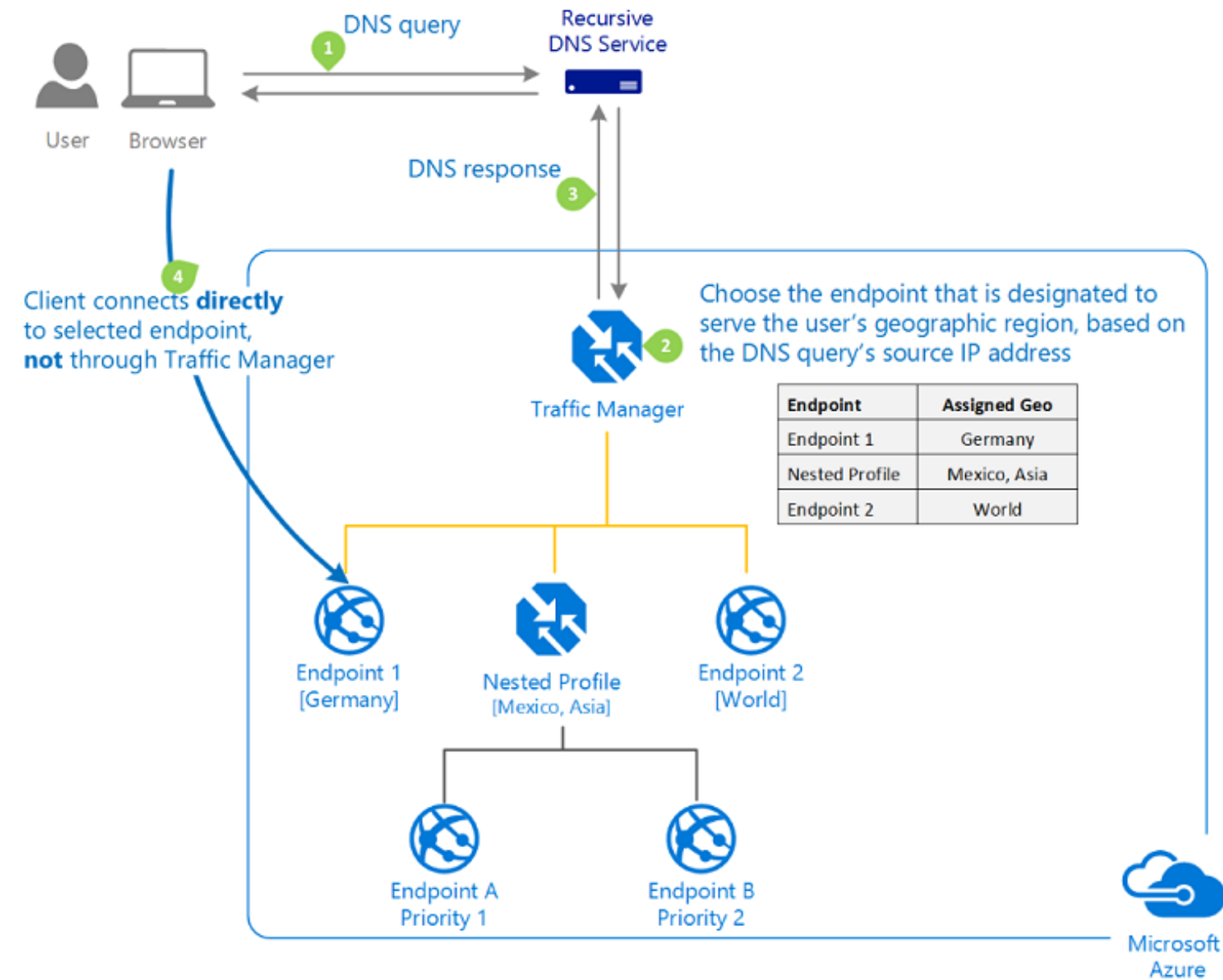
This method helps users route traffic to their nearest location.



Source: <https://learn.microsoft.com/en-us/azure/traffic-manager/traffic-manager-routing-methods>

Geographic Routing

This method helps in measuring traffic from different regions.



Source: <https://learn.microsoft.com/en-us/azure/traffic-manager/traffic-manager-routing-methods>

Points to Remember

All Azure Traffic Manager profiles use the shared domain ***.trafficmanager.net**.

Web apps in the same data centers will automatically be load-balanced.

Web apps spread across multiple data centers will need to be load-balanced.

Endpoints need to be in the same subscription to provide load balancing between the apps.

Distributing Network Traffic

The network traffic table is explained below:

Service	Azure Load Balancer	Application Gateway	Traffic Manager	Azure Front Door
Technology	Transport layer (Level 4)	Transport layer (Layer 7)	DNS resolver	Layer 7 or HTTP/HTTPS
Protocols	Any TCP or UDP protocol	HTTP, HTTPS, HTTP/2, and Web sockets	DNS resolution	Split TCP-based anycast protocol
Backends and endpoints	Azure VMs and Azure VM scale sets	Azure VMs, Azure VM scale sets, Azure app services, IP addresses, and hostnames	Azure Cloud Services, Azure App Services, Azure App Services slots, and public IP addresses	Internet-facing services hosted inside or outside of Azure
Network connectivity	External and Internal	External and Internal	External	External and Internal

Creating Azure Traffic Manager



Duration: 10 Min.

Problem Statement:

Sim-Edu Corporation wants to use Azure Traffic Manager and wants to see its working functionality by checking how high availability works with Azure Traffic Manager. You have been assigned the task of implementing Traffic Manager, which will route client requests to the proper service endpoint using a traffic-routing method.

Outcome:

You will be able to demonstrate the implementation of Azure Traffic Manager to ensure high availability by effectively routing client requests across multiple endpoints using a chosen traffic-routing method, showcasing enhanced service continuity and reliability.

Note: Refer to the demo document for detailed steps:
[04_Creating_Azure_Traffic_Manager](#)

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed are:

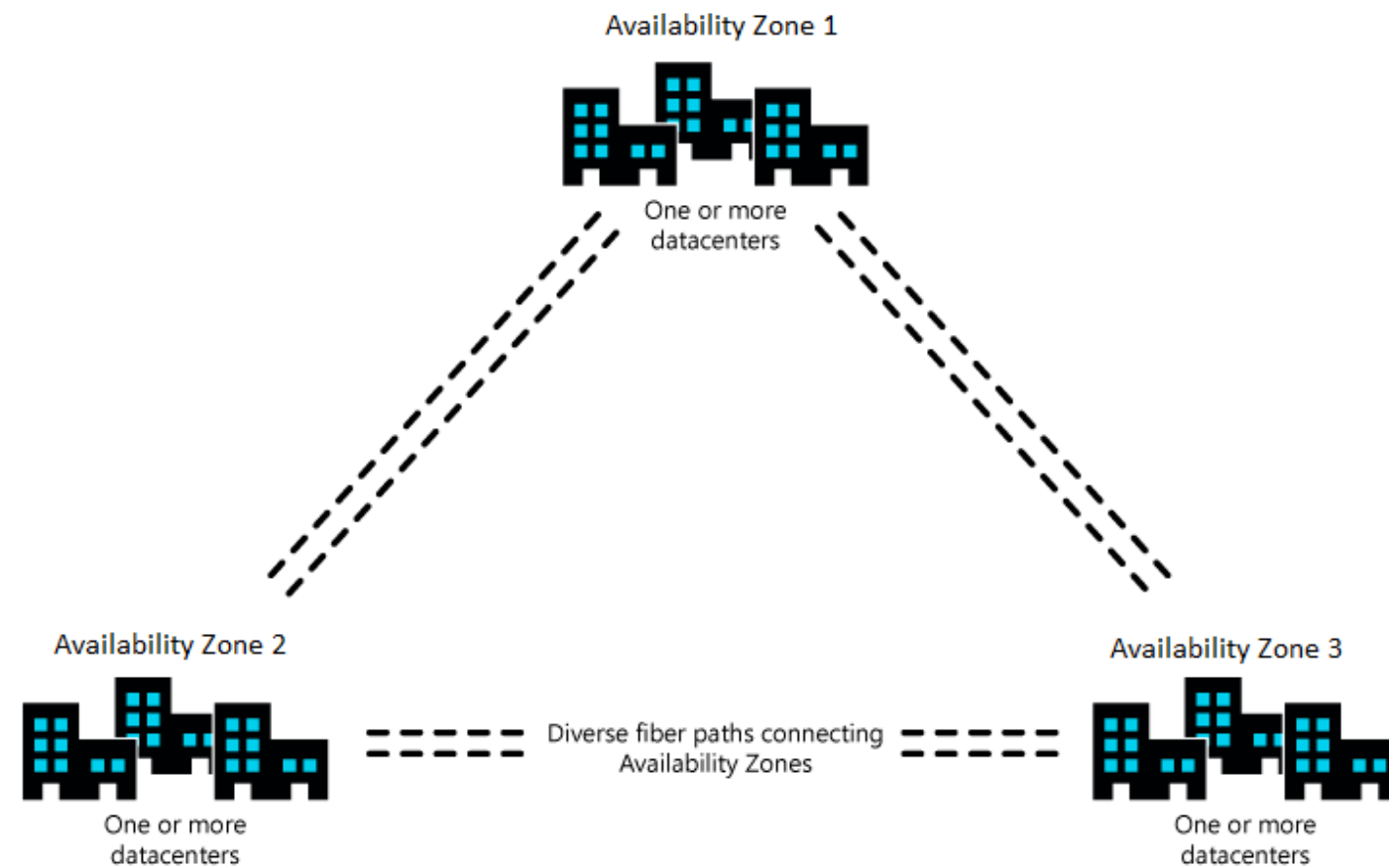
1. Create virtual machines
2. Add DNS names to virtual machines
3. Create Traffic Manager
4. Test Traffic Manager



Recommend a High Availability Solution for Compute

Building Solutions Using Availability Zones

Availability Zones are physical locations inside an Azure region that are distinct from one another.

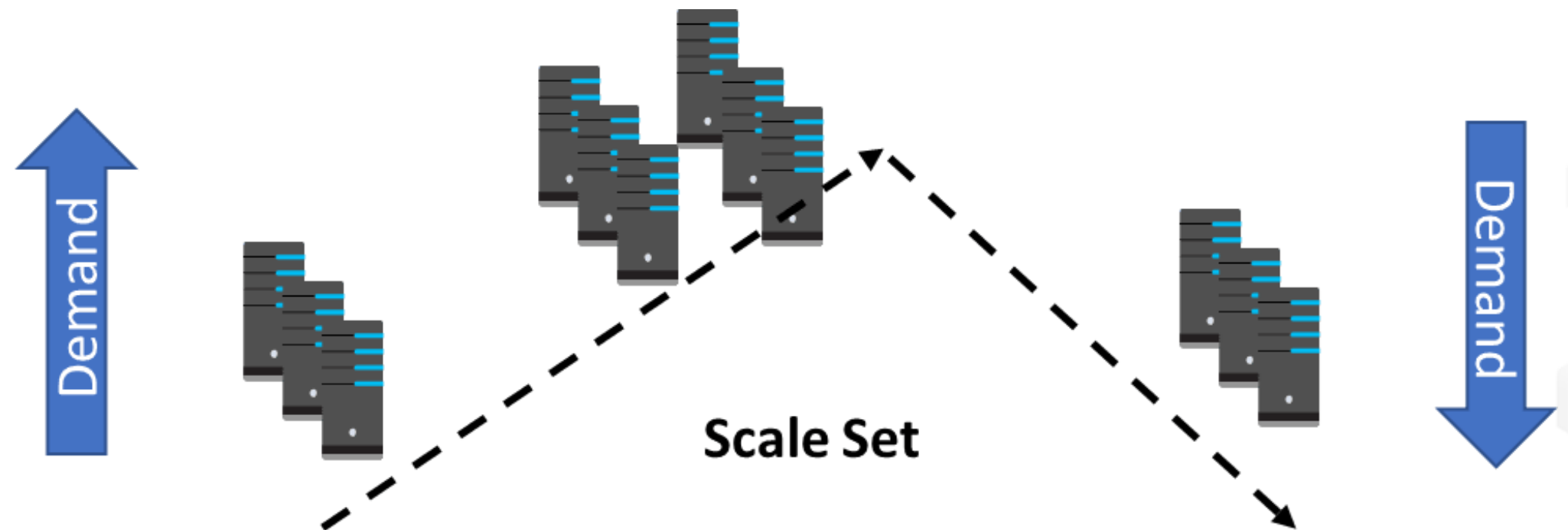


- A zone is made up of one or more data centers with independent power, cooling, and networking.
- The physical separation of availability zones within a region limits the impact on applications.
- Availability zones ensure that your data remains protected even if there's an event impacting a datacenter, such as a power outage or cooling issues.

Zones are designed to support services, capacity, and availability from other zones in the region.

Virtual Machine Scale Sets

It enables the user to deploy a set of identical VMs.



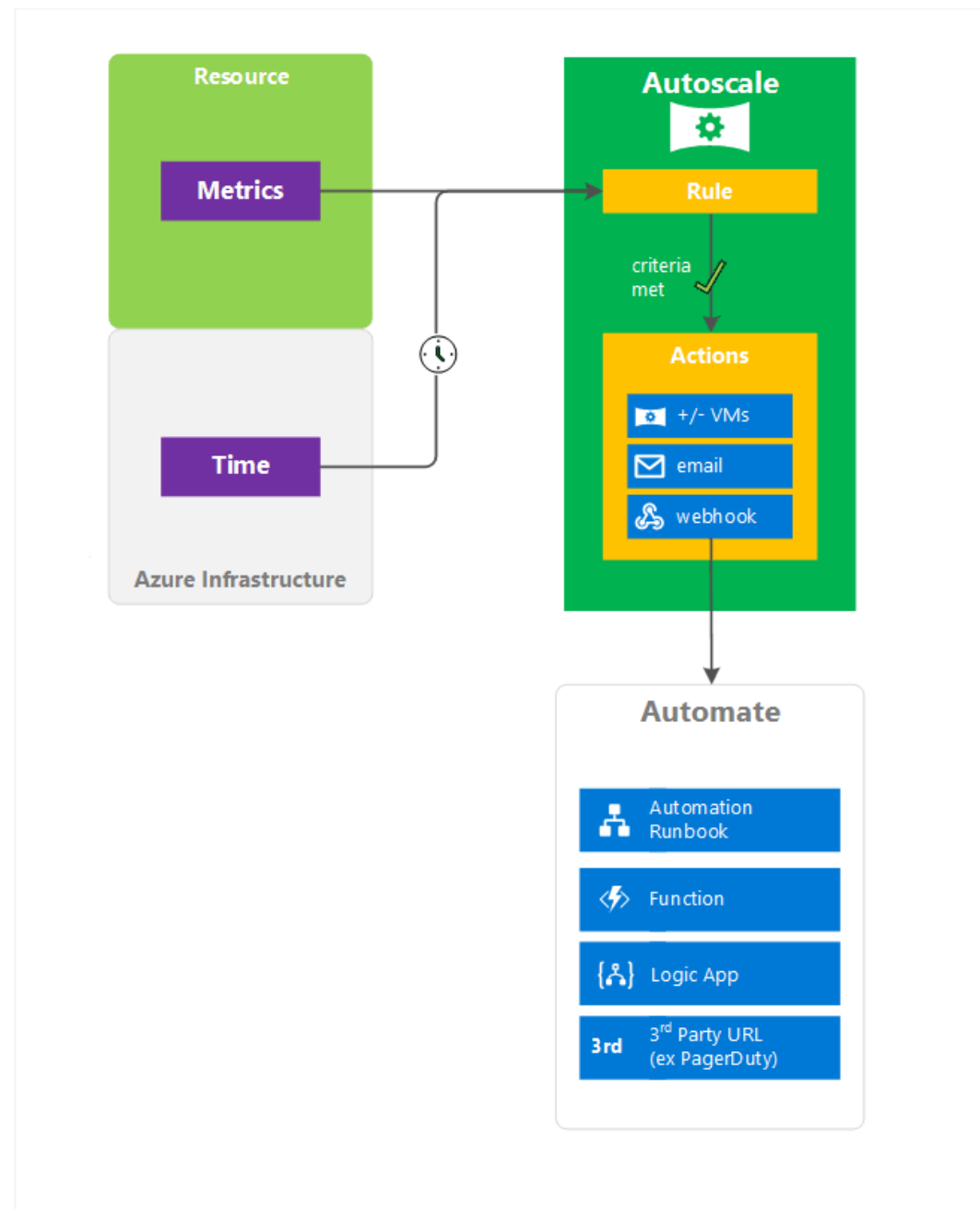
The number of VM instances can automatically increase or decrease in response to demand or a defined schedule.

Virtual Machine Scale Sets: Benefits

- Simplifies the creation and management of multiple VMs
- Provides high availability and application resiliency by distributing VMs across availability zones or fault domains
- Allows your application to automatically scale as resource demand changes
- Works on a large scale



Autoscale



- It is a service that automatically adds and removes resources according to the load on the application.
- It adjusts resources dynamically, adding them during high loads and reducing them during low loads to optimize costs.

Source: <https://learn.microsoft.com/en-us/azure/azure-monitor/autoscale/autoscale-overview>

Autoscale

Create virtual machine scale set

AUTOSCALE

Autoscale ⓘ Disabled Enabled

* Minimum number of VMs ⓘ

* Maximum number of VMs ⓘ

Scale out

* CPU threshold (%) ⓘ

* Number of VMs to increase by ⓘ

Scale in

* CPU threshold (%) ⓘ

* Number of VMs to decrease by ⓘ

- Defines the minimum, maximum, and default numbers of VM instances
- Creates more advanced scale sets with scale-out and scale-in parameters

Autoscale Conditions

The following are the autoscale conditions can be set to manage scaling in the infrastructure:

Percentage CPU
or load

Scale based on the percentage of CPU usage or system load

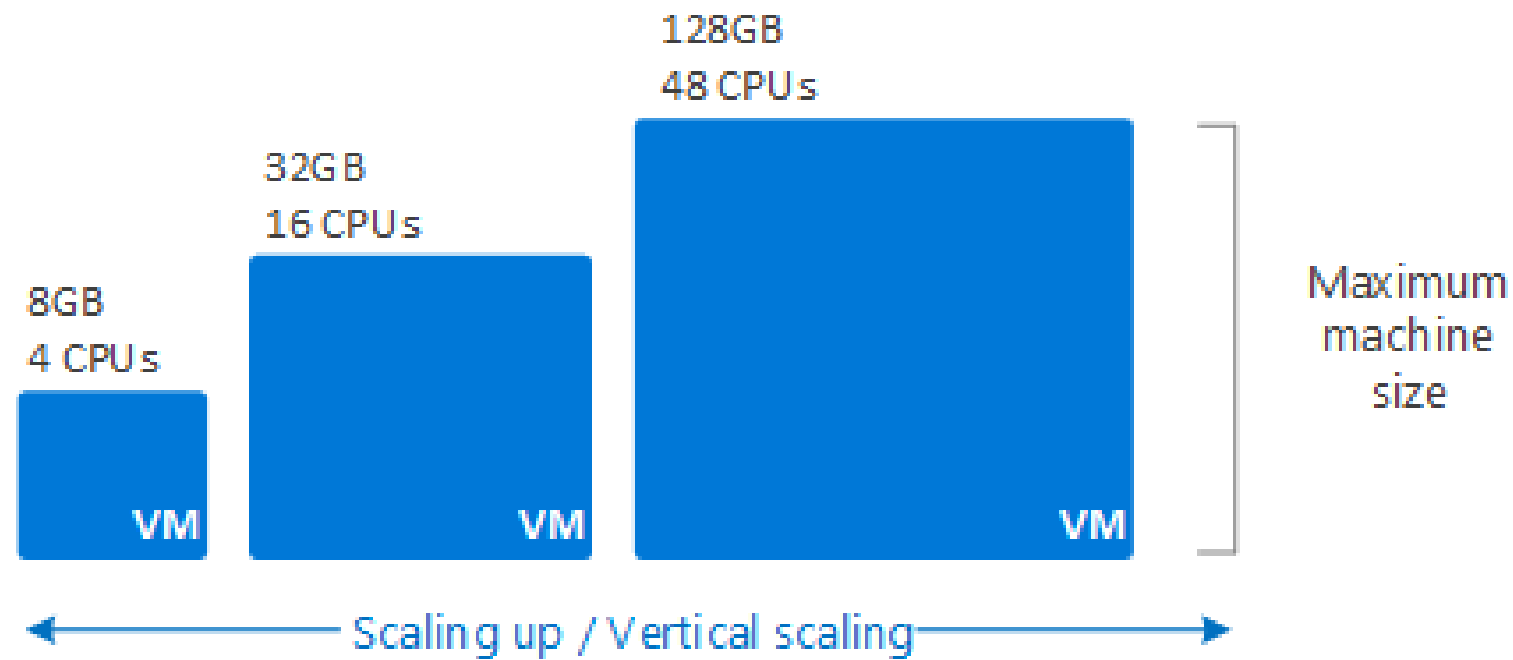
Scheduled scaling

Automatically scale up or down based on a schedule, such as every Sunday

Predictive
autoscale

Predicts future scaling needs and optimizes scaling policies or rules accordingly

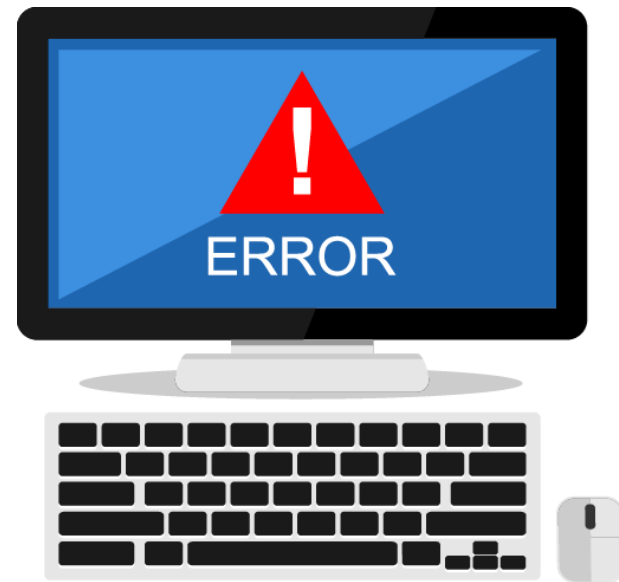
Horizontal vs. Vertical scaling



- Horizontal scaling involves adjusting the number of resource instances, either increasing or decreasing.
- Vertical scaling, or scaling up and down, maintains a consistent number of resource instances while boosting their capacity in memory, CPU speed, disk space, and network.

Azure Storage Redundancy

Azure Storage stores multiple copies of user data to protect from planned and unplanned events, such as:



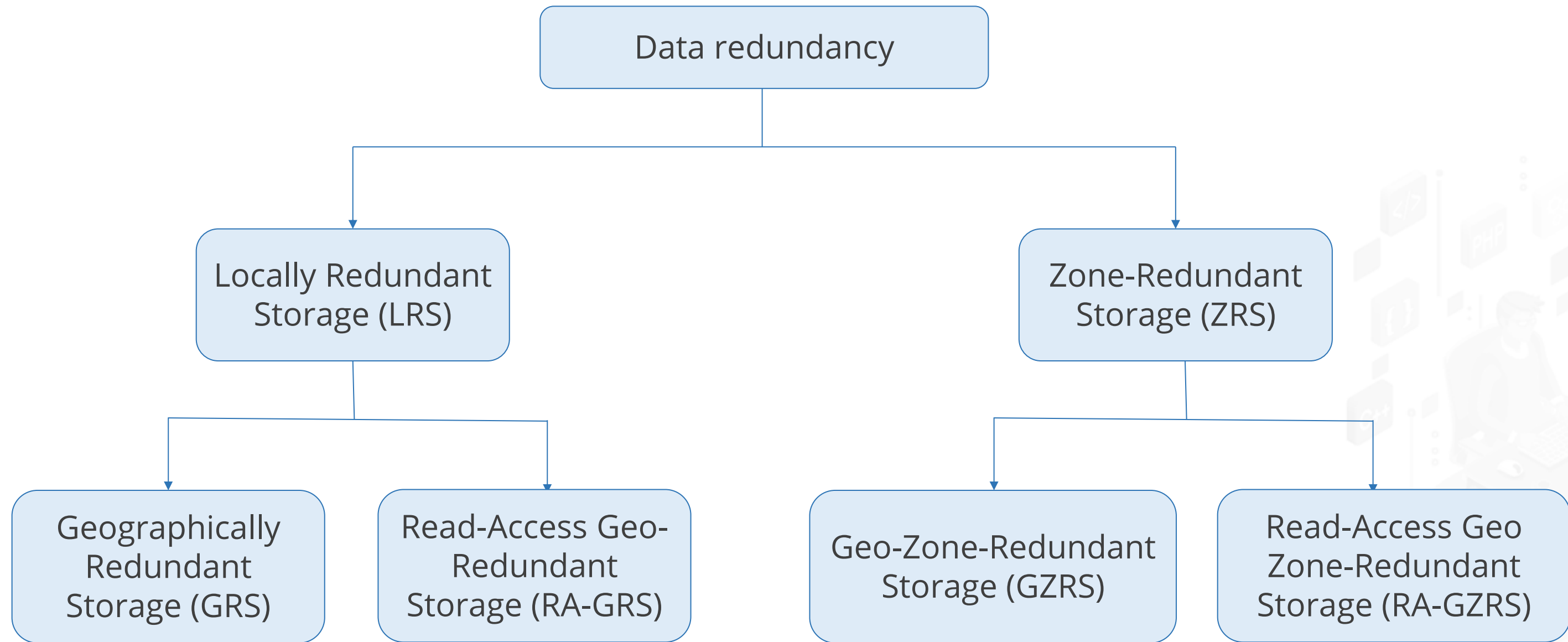
Hardware failures

Network or power outages

Massive natural disasters

Redundancy ensures that a storage account meets its availability and durability targets even in the face of failures.

Azure Storage Redundancy



Data in an Azure Storage account is replicated three times in the primary region.

Setting Up and Configuring Azure Virtual Machine Scale Sets



Duration: 10 Min.

Problem Statement:

You were tasked with exploring deployment and configuration options for Azure virtual machines to enhance availability using VM Scale Sets. This includes examining compute and storage scalability and resiliency, as well as automating configurations using the Azure Virtual Machine Custom Script extension.

Outcome:

You will be able to demonstrate how to deploy and configure Azure virtual machines with enhanced availability using VM Scale Sets and automate their setup using the Azure Virtual Machine Custom Script extension.

Note: Refer to the demo document for detailed steps:
[05_Setting_Up_and_Configuring_Azure_Virtual_Machine_Scale_Sets](#)

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed are:

1. Register the Microsoft resource providers
2. Deploy zone-resilient Azure Virtual Machine Scale Sets by using the Azure portal
3. Configure Azure Virtual Machine Scale Sets by using virtual machine extensions



Recommend a High Availability Solution for Relational Data Storage

Hyperscale Availability Model

The availability model in Hyperscale includes four layers:

Stateless compute layer

Includes the primary compute replica and, optionally, several secondary compute replicas

Stateless storage layer

Serves as the distributed storage engine for database engine processes on compute replicas

Hyperscale Availability Model

The availability model in Hyperscale includes four layers:

Stateful transaction log storage layer

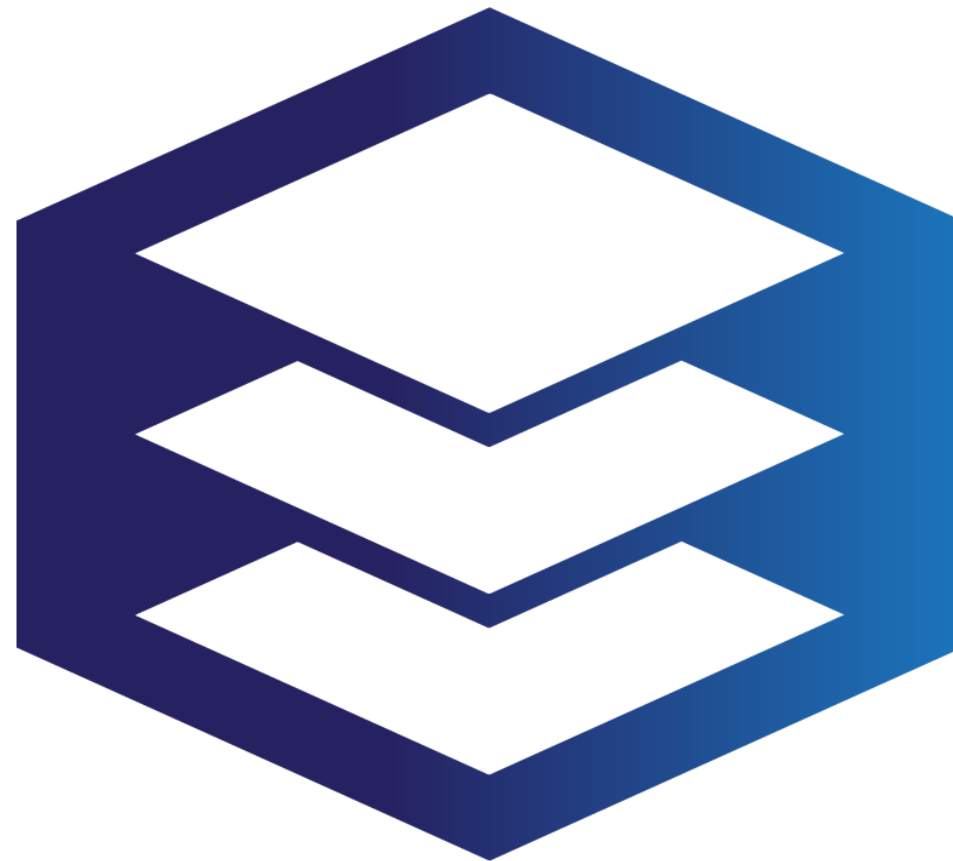
Provides availability and redundancy for transaction logs, ensuring data durability for committed transactions

Stateful data storage layer

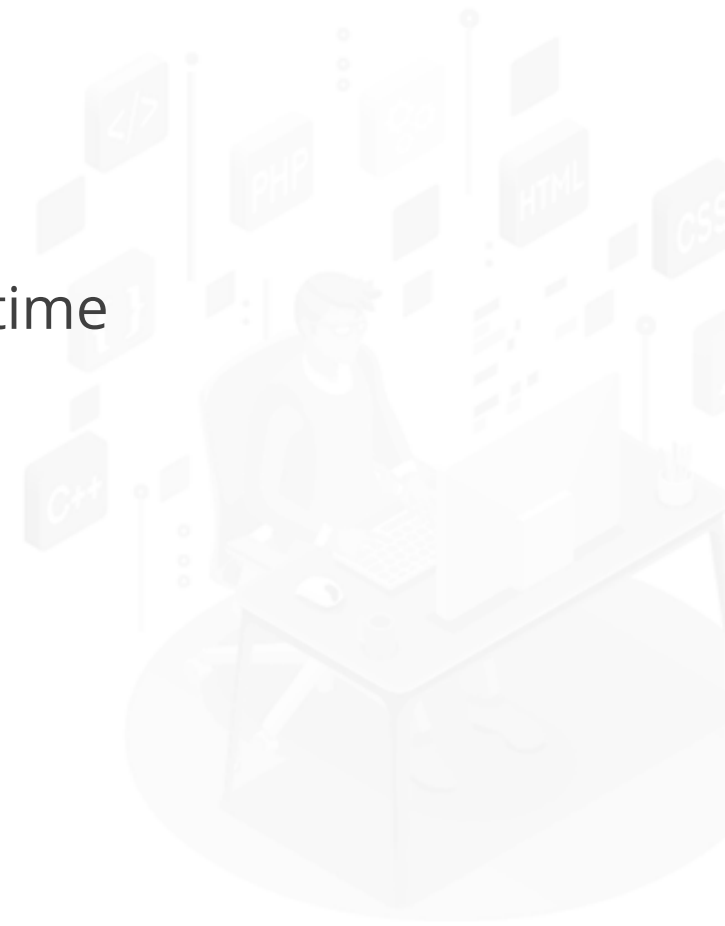
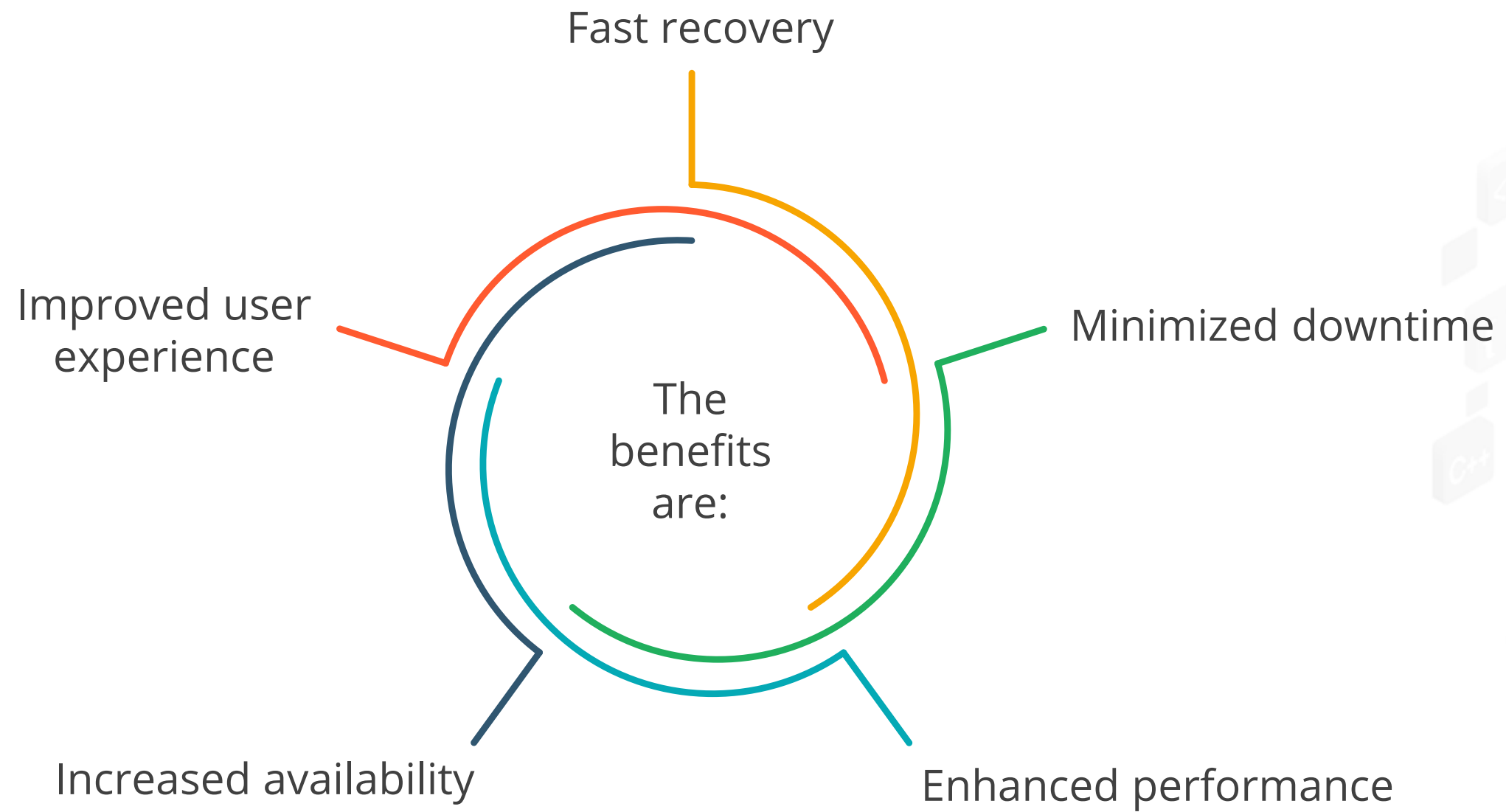
Ensures the preservation of every page in a data file during crashes in other layers or compute node failures within the Hyperscale architecture

Accelerated Database Recovery (ADR)

It improves database availability, especially during long-running transactions, by redesigning the SQL database engine recovery process.

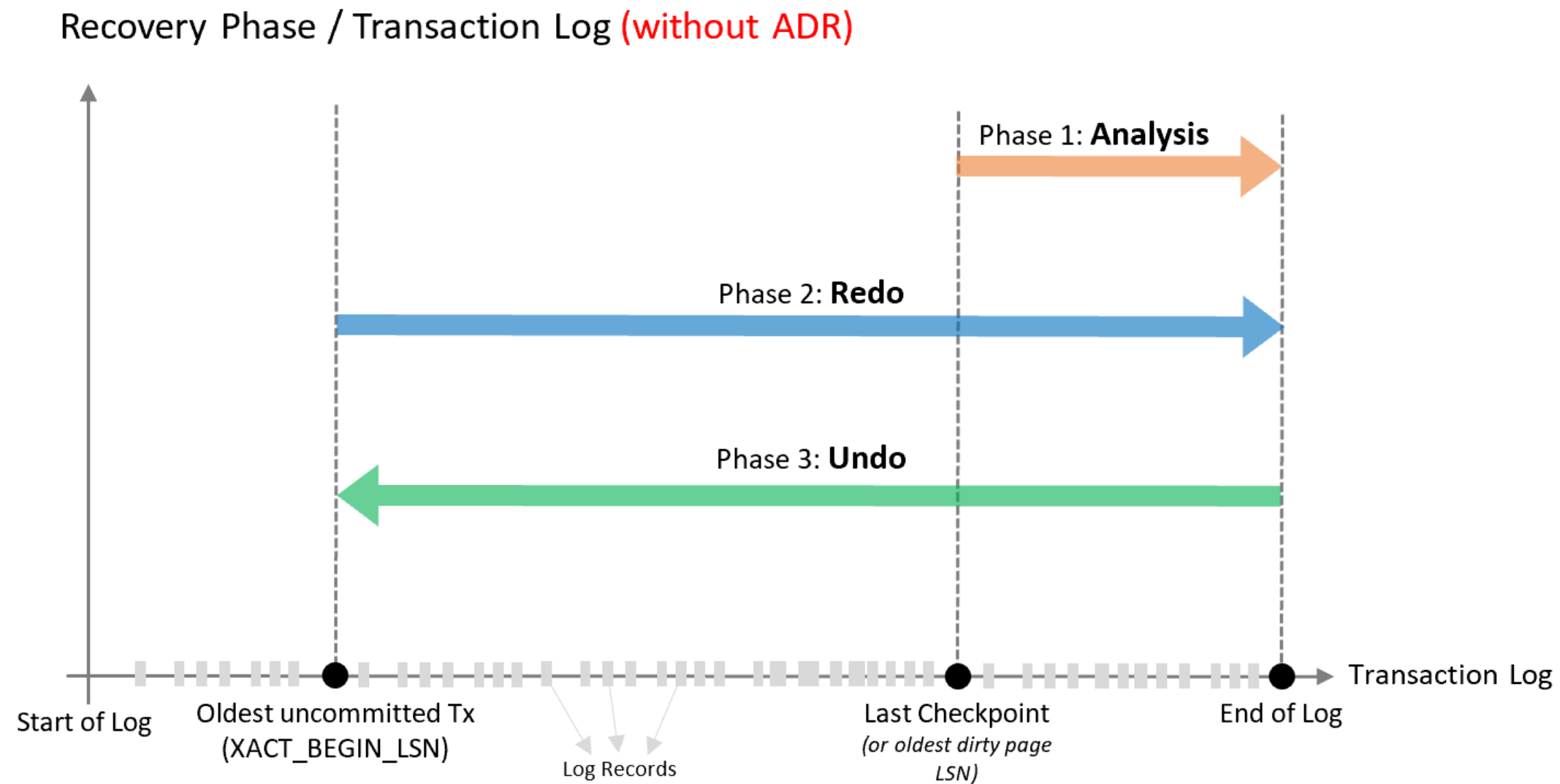


ADR



Standard Database Recovery Process

It follows the ARIES recovery model and consists of three phases, which are illustrated in the following diagram:



Standard Database Recovery Process



Analysis phase

Identifies the status of each transaction at the time the database was shut down

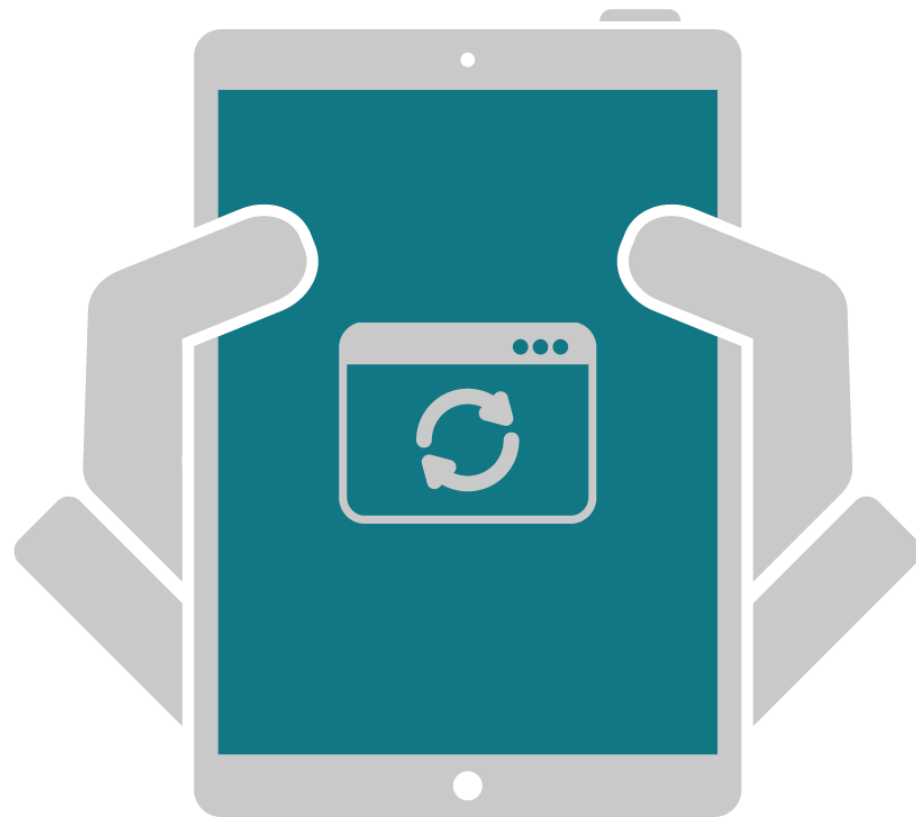
Standard Database Recovery Process



Redo phase

Restores the database to its state before the accident by undoing all committed operations

Standard Database Recovery Process

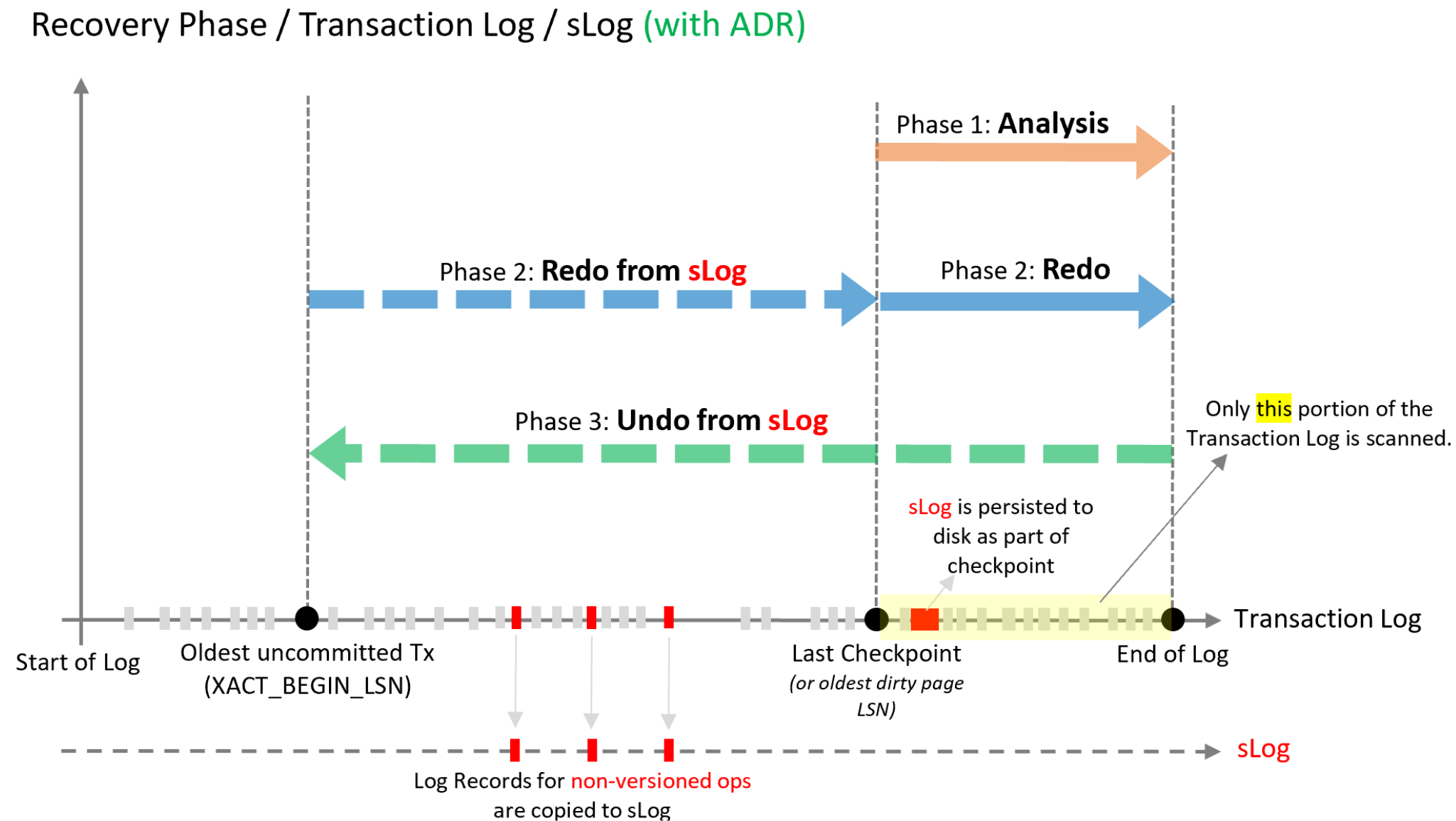


Undo phase

Traverses the log backward for each transaction that was active at the time of the crash, redoing the actions that this transaction executed

Accelerated Database Recovery Process

It enables swift database recovery by versioning physical modifications and instantly undoing limited logical operations.



Accelerated Database Recovery Process



Analysis phase

The process remains the same as before with the addition of reconstructing sLog and copying log records for non-versioned operations.

Accelerated Database Recovery Process



Redo phase

Subphase 1:

Redo from SLOG (oldest uncommitted transaction up to the last checkpoint)

Subphase 2:

Redo from the transaction log starts from the last checkpoint (instead of the oldest uncommitted transaction)

Accelerated Database Recovery Process



Undo phase

It completes instantaneously using sLog to undo non-versioned operations and persisted version store (PVS) with logical revert to perform row-level version-based undo.

Accelerated Database Recovery

Persisted Version Store (PVS):

It is a new SQL Server database engine.



sLog:

It is a secondary in-memory log stream that stores log records for non-versioned operations.

Logical revert:

It is an asynchronous mechanism that performs row-level version-based undo.

Cleaner:

It is an asynchronous operation that runs regularly and removes outdated page versions.

Accelerated Database Recovery

The workloads listed below benefit the most from ADR:

- Workloads with long-running transactions
- Workloads that have seen cases where active transactions are causing the transaction log to grow significantly
- Workloads that have experienced long periods of database unavailability due to a long-running recovery

06 Creating Azure SQL DB Geo Replication



Duration: 10 Min.

Problem Statement:

You have been asked to demonstrate the use Azure SQL DB Geo-Replication to construct readable secondary databases of individual databases on a server in the same or a different data center.

Outcome:

You will be able to demonstrate how to use Azure SQL DB Geo-Replication to set up readable secondary databases across different data centers, enhancing data availability and disaster recovery capabilities.

Note: Refer to the demo document for detailed steps:
[06_Creating_Azure_SQL_DB_Geo_Replication](#)

ASSISTED PRACTICE

Assisted Practice: Guidelines

Steps to be followed are:

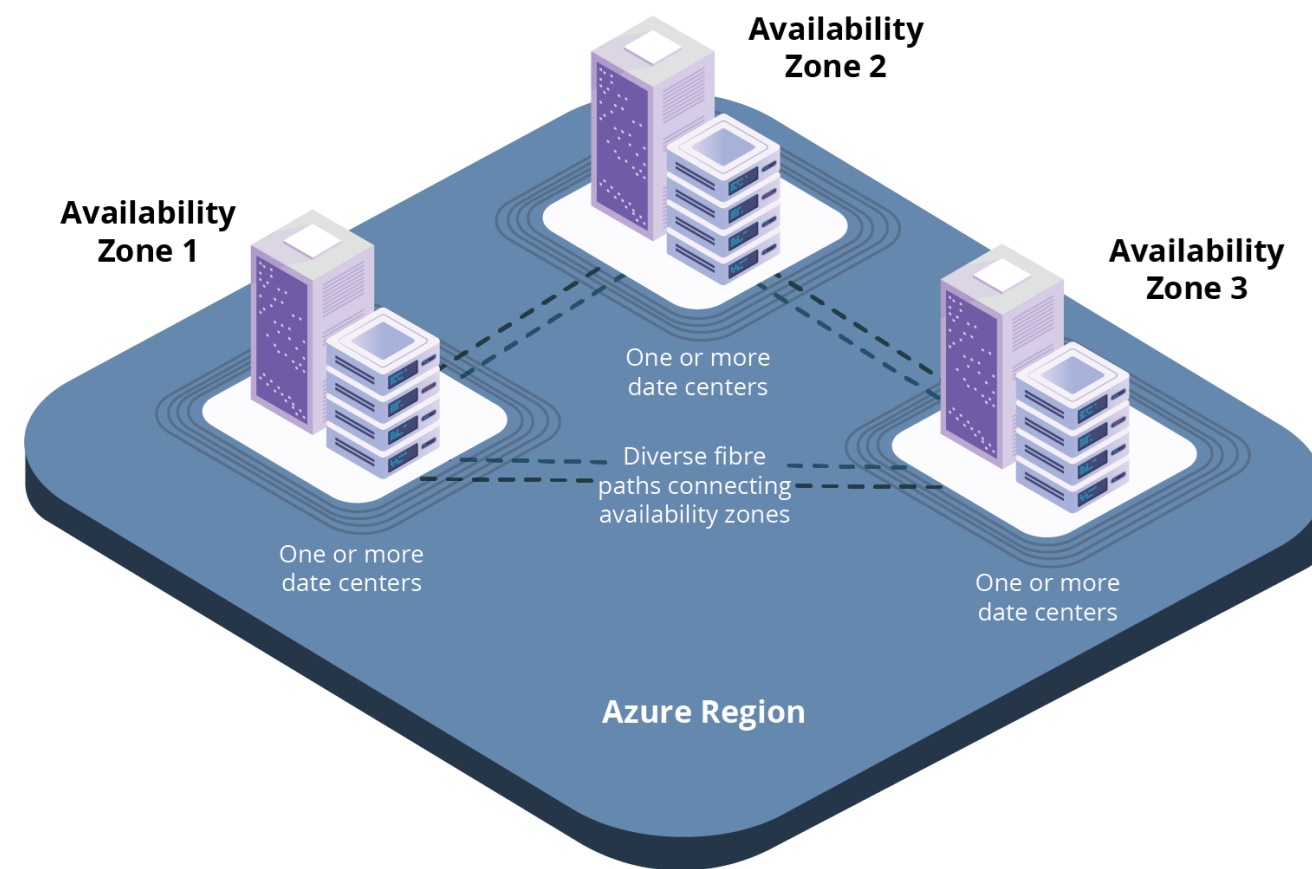
1. Create Azure SQL database
2. Create a secondary database using the geo-replication



Recommend a High Availability Solution for Non-Relational Data

High Availability Using Availability Zones

Azure regions and availability zones are meant to assist users in achieving resilience and reliability for their goal workloads.



Datacenters are placed within a latency-defined perimeter in each Azure region.

Delivering Reliability in Azure

The key to improving the reliability of a solution is to design it to continue to function despite failure.

In cloud-based solutions, building to survive failure is a shared responsibility. These responsibilities can be viewed at three levels, such as:

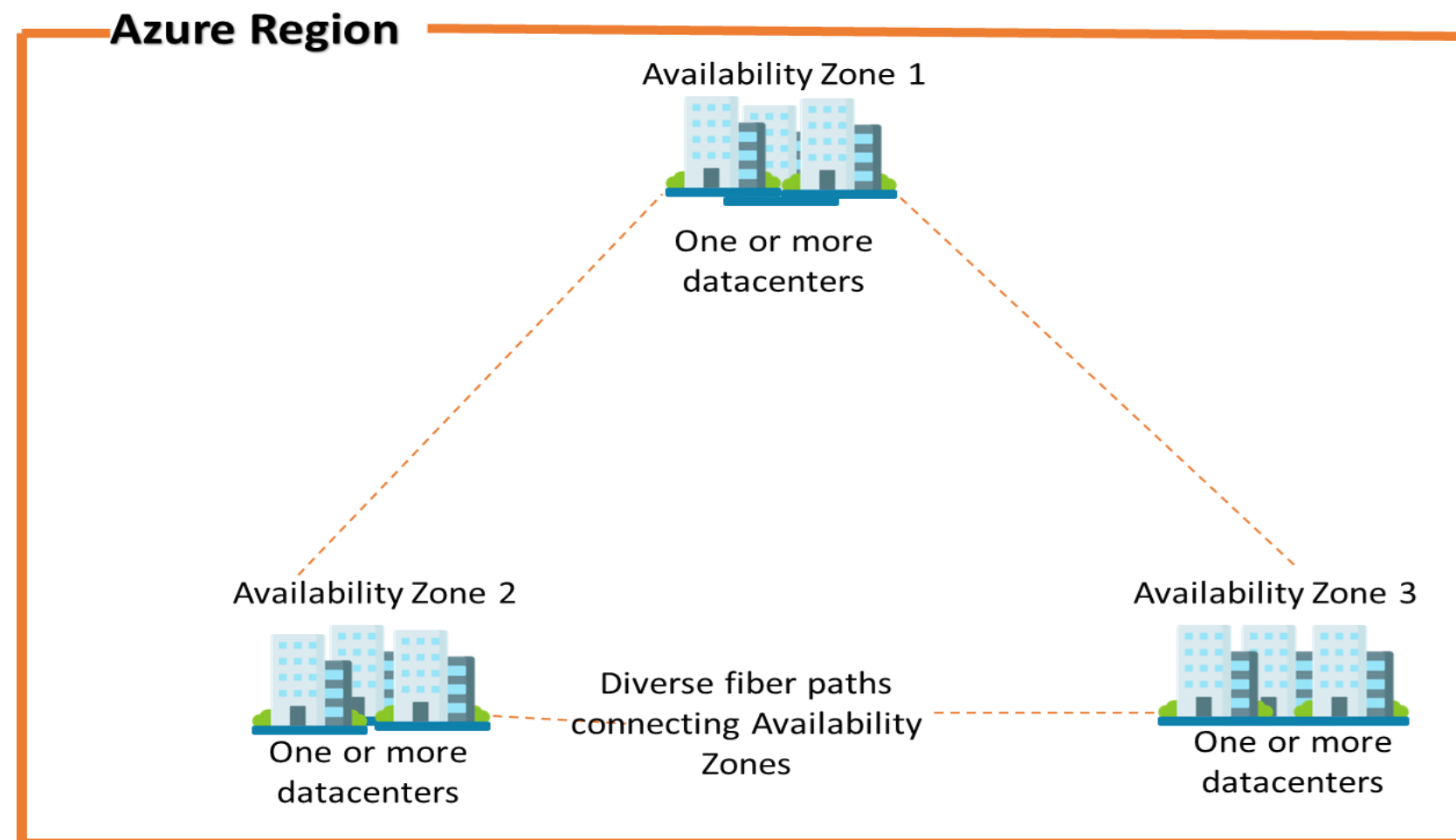
Resilient foundation

Resilient applications

Resilient services

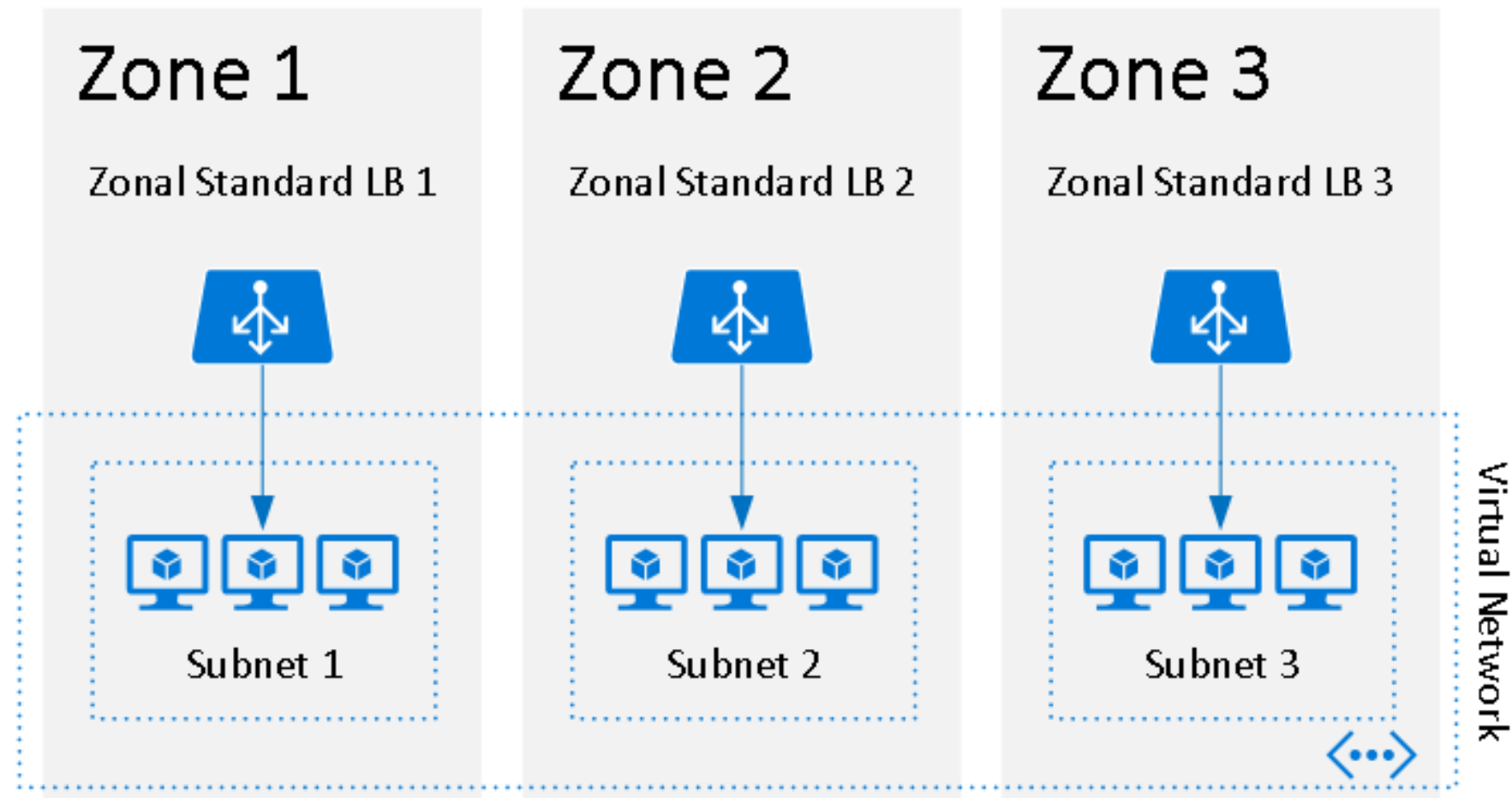
Zonal and Zone-Redundant Architecture

Azure services supporting availability zones fall into two categories: zonal and zone-redundant.



Zonal Architecture

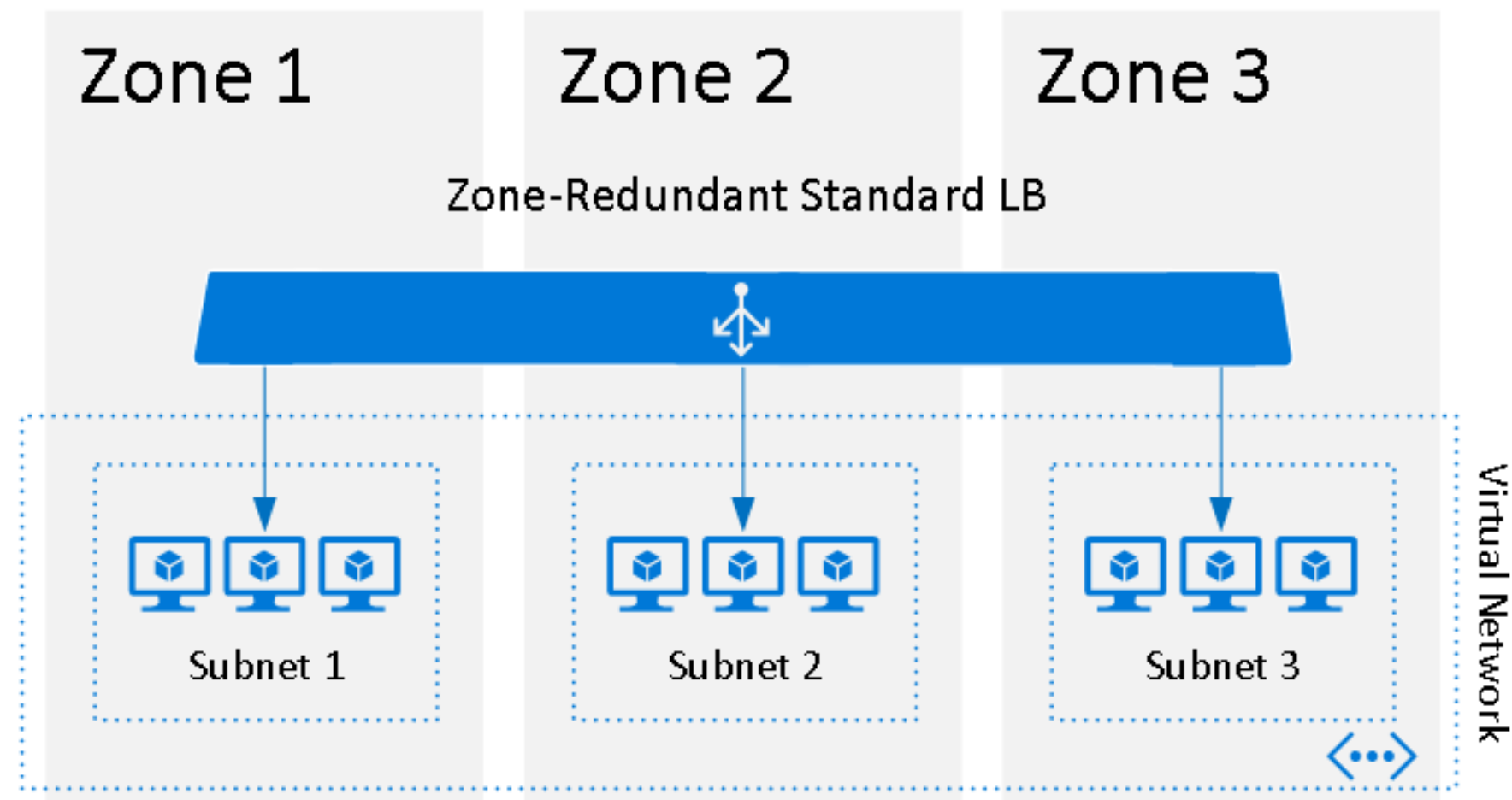
It helps achieve strict latency or performance requirements by enabling the pinning of multiple resources to a specific zone, ensuring traffic between them need not travel far.



Source: <https://learn.microsoft.com/en-us/azure/architecture/high-availability/building-solutions-for-high-availability>

Zone-Redundant Architecture

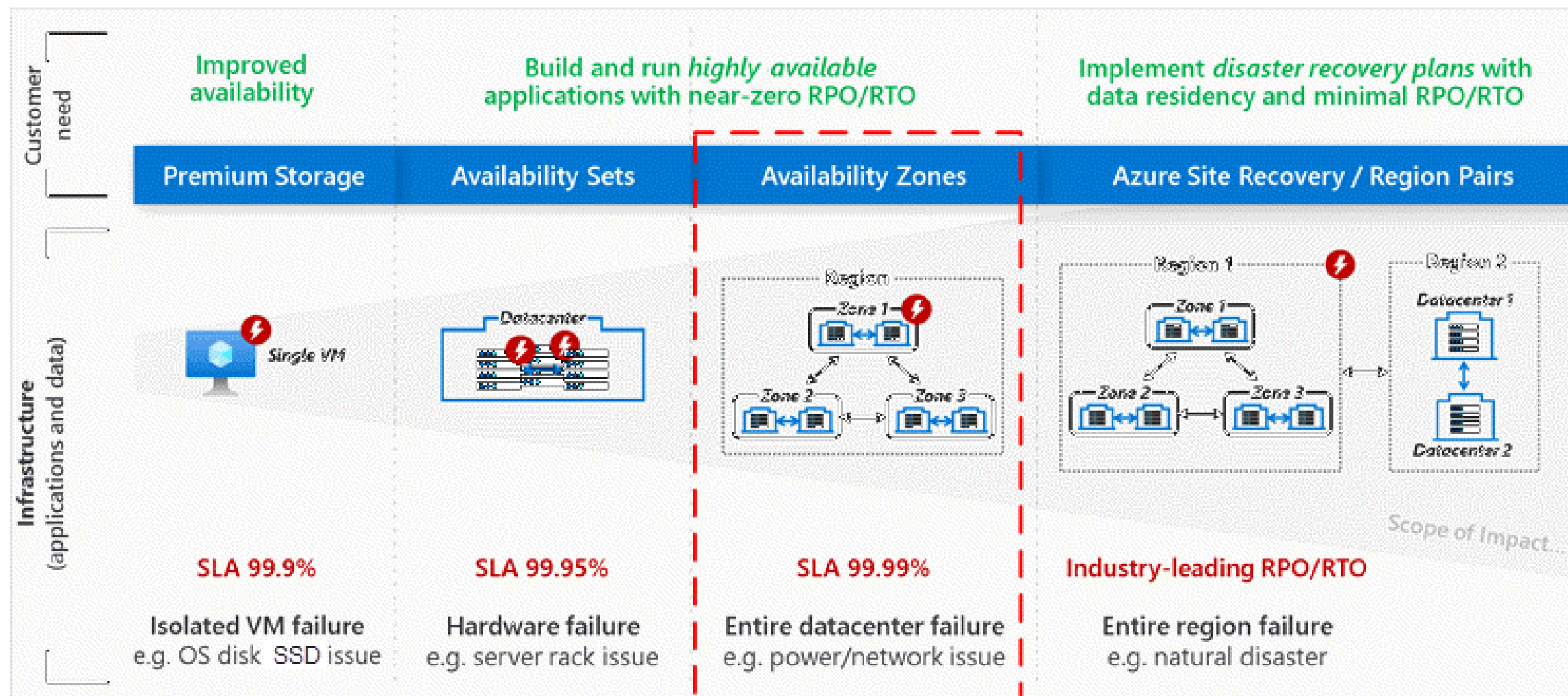
It replicates a resource and its configuration and data across multiple availability zones.



Source: <https://learn.microsoft.com/en-us/azure/architecture/high-availability/building-solutions-for-high-availability>

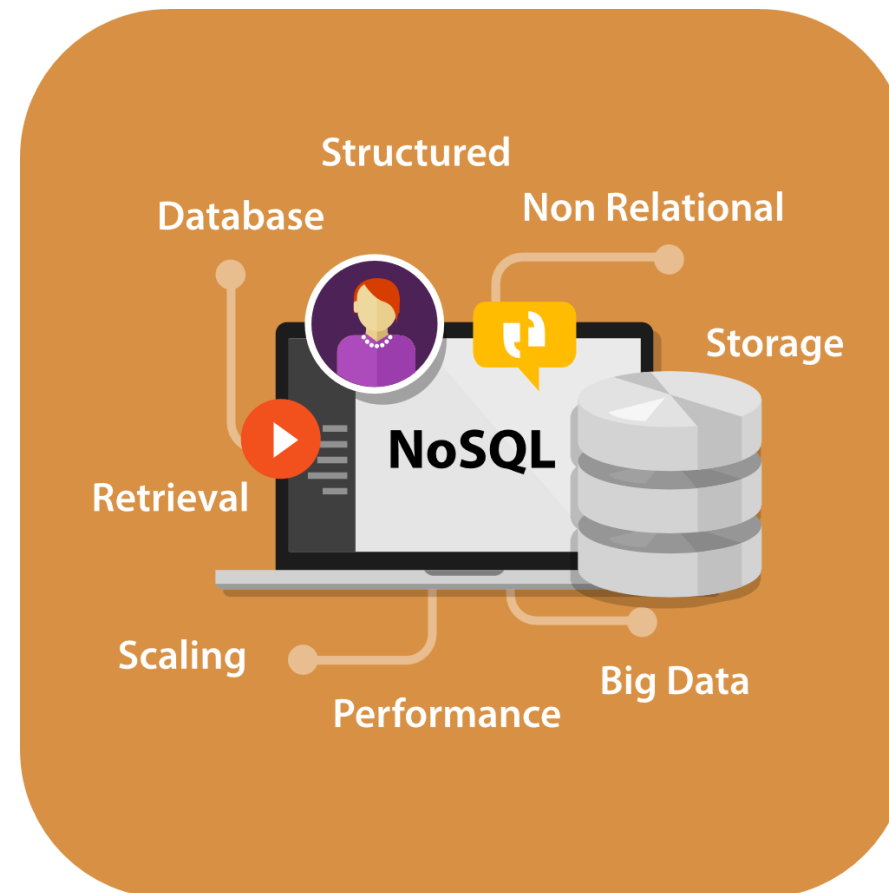
SLA Offered by Availability Zones

The following diagram illustrates the different levels of HA offered by a single VM, availability sets, and availability zones:



Relational and NoSQL Data

These are two types of database systems commonly implemented in cloud-native apps.



They are designed to store data in different ways and can be accessed in multiple ways.



Relational and NoSQL Data

Relational database

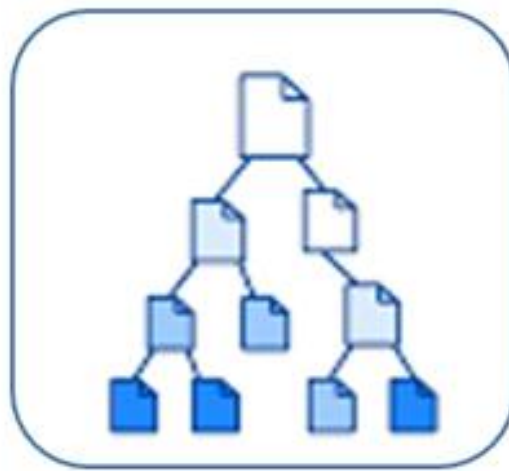
Offers a repository of interconnected data tables with a fixed schema, utilizing SQL (Structured Query Language) for data management and providing support for ACID guarantees

No SQL database

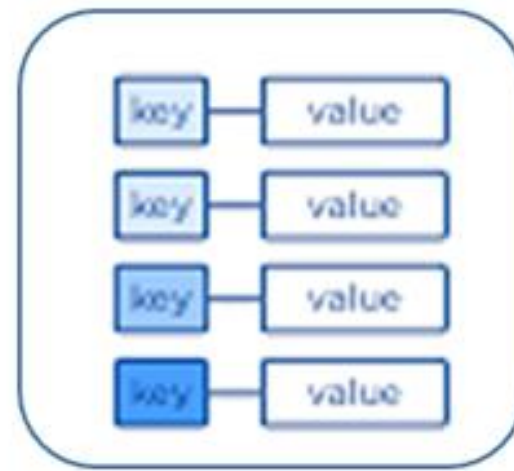
Refers to high-performance, non-relational data stores known for excelling in ease-of-use, scalability, resilience, and availability characteristics

NoSQL Databases: Data Models

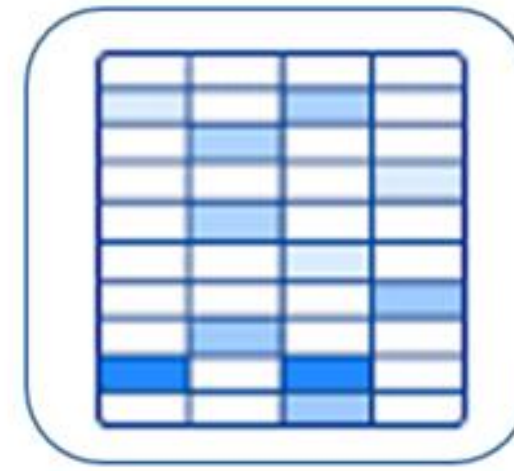
NoSQL databases include several different models for accessing and managing data, each suited to specific use cases. The four data models are:



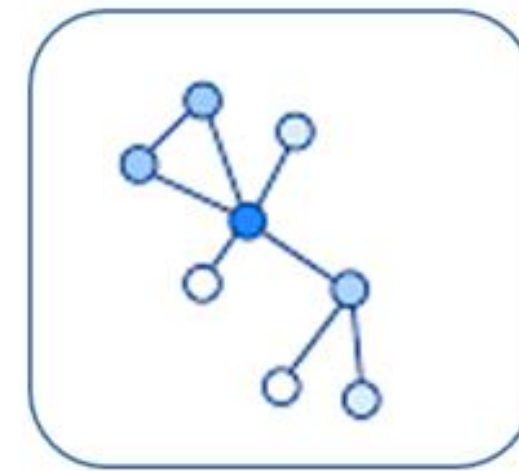
Document
Store



Key-Value
Store



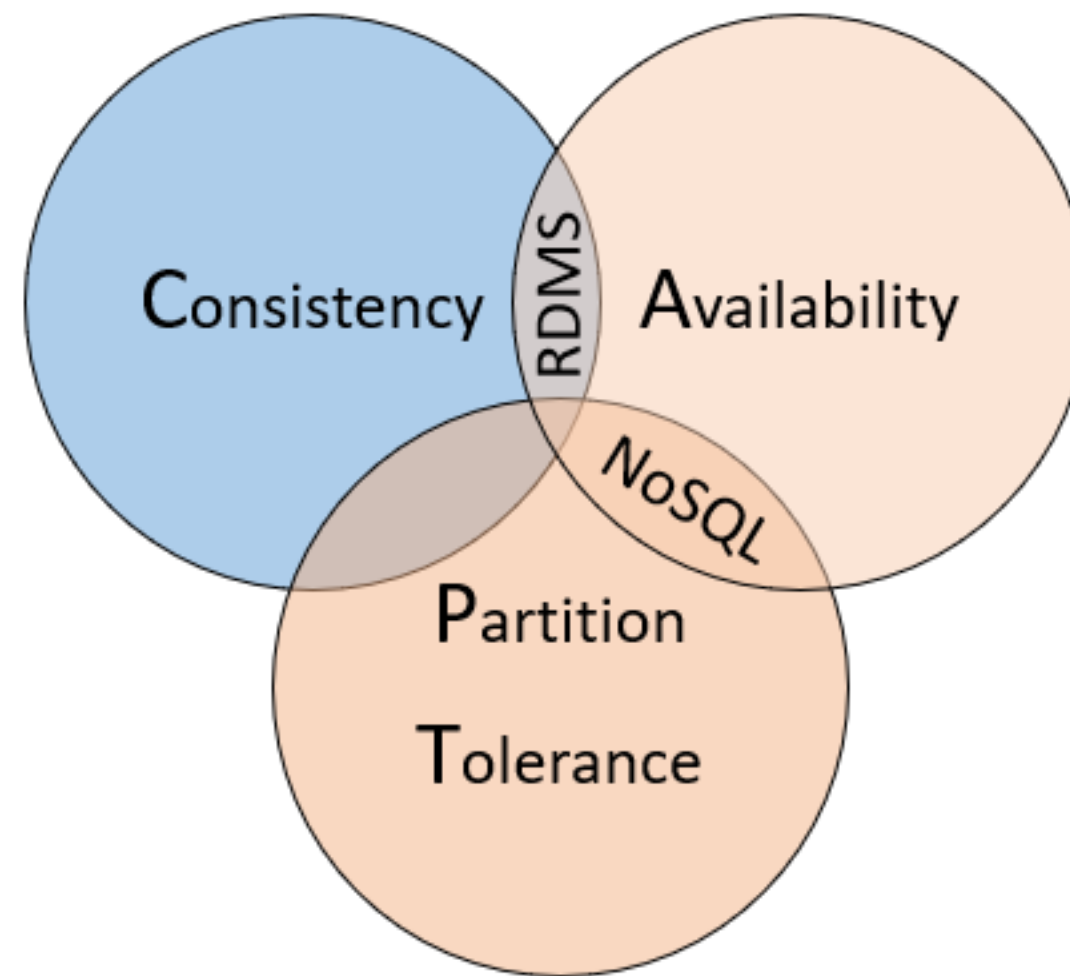
Wide-Column
Store



Graph
Store

The CAP Theorem

It states that distributed data systems offer a trade-off between consistency, availability, and partition tolerance.



The CAP Theorem: Properties

Availability

Every node returns an immediate response, even if that response isn't the most recent data.

Consistency

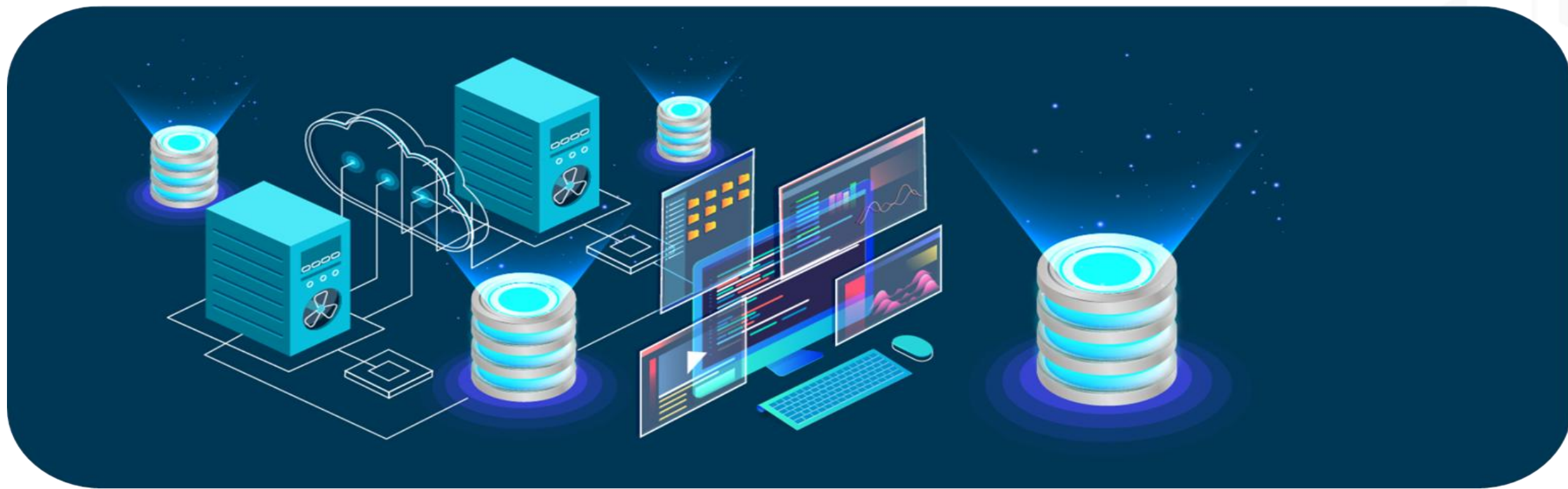
Every node in the cluster responds with the most recent data, even if the system must block the request until all replicas update.

Partition tolerance

It ensures system continuity despite a replicated data node failure or loss of connectivity with other nodes.

The CAP Theorem: Properties

Users may set up, run, and scale databases using a common set of abstractions (primitives) without knowing or caring about the actual implementations.



DBaaS on IaaS

The DBaaS solution would request resources from the underlying IaaS, which would handle the provisioning of computing, storage, and networking as needed, and effectively eliminate the need for IT involvement.



Benefits of DBaaS

Developer agility

IT productivity

Application reliability and performance

Application security



Azure Relational Databases

Azure provides four managed relational DBaaS packages for cloud-native microservices that require relational data, as shown in the table below:

SQL Database	My SQL	MariaDB	PostgreSQL
Shared Database Services Platform	•Built-in high availability with a 99.99% service level agreement •Automatic upgrades, patching, and backups •On-demand scaling with pay-as-you-go pricing •Intelligence: Advisors, tuning, and monitoring •Active geo-replication with readable secondary databases •Enterprise-grade security to protect data at rest and in motion		
Azure Compute layer			
Azure Storage layer			
Globally available across 54 Azure regions			

Azure SQL Database

It is a Microsoft SQL Server Database Engine-based, fully managed relational DBaaS.



The service shares many features found in the on-premises version of SQL Server and runs the latest stable version of the SQL Server Database Engine.

Deployment Options

1

A Single Database represents a fully managed SQL Database running on an Azure SQL Database server in the Azure cloud.

2

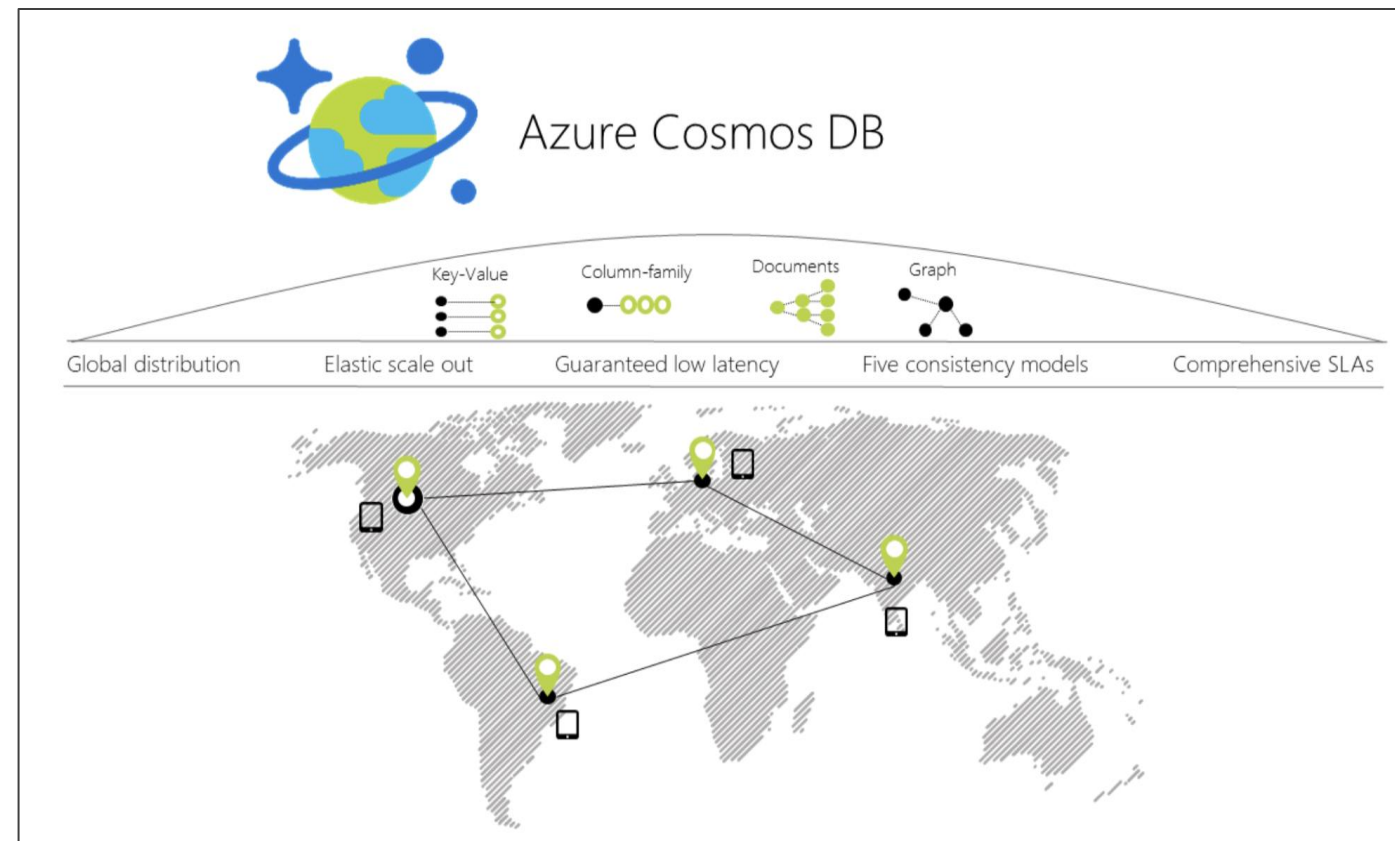
A Managed Instance is a fully managed instance of the Microsoft SQL Server Database Engine that provides nearly 100 percent compatibility.

3

Serverless computing in Azure SQL Database is a compute tier for a single database that scales automatically based on workload requirements.

NoSQL Solutions with Azure Cosmos DB

Azure Cosmos DB is a fully managed NoSQL database service that offers global distribution, automatic scaling, and low latency, supporting multiple data models such as document, key-value, graph, and column-family.



Identify Storage Types for High Availability

Storage Account Failover

It prevents unplanned service outages for the user.



- It is an integral part of the disaster recovery plan.
- It helps the user if the primary endpoint becomes unavailable.

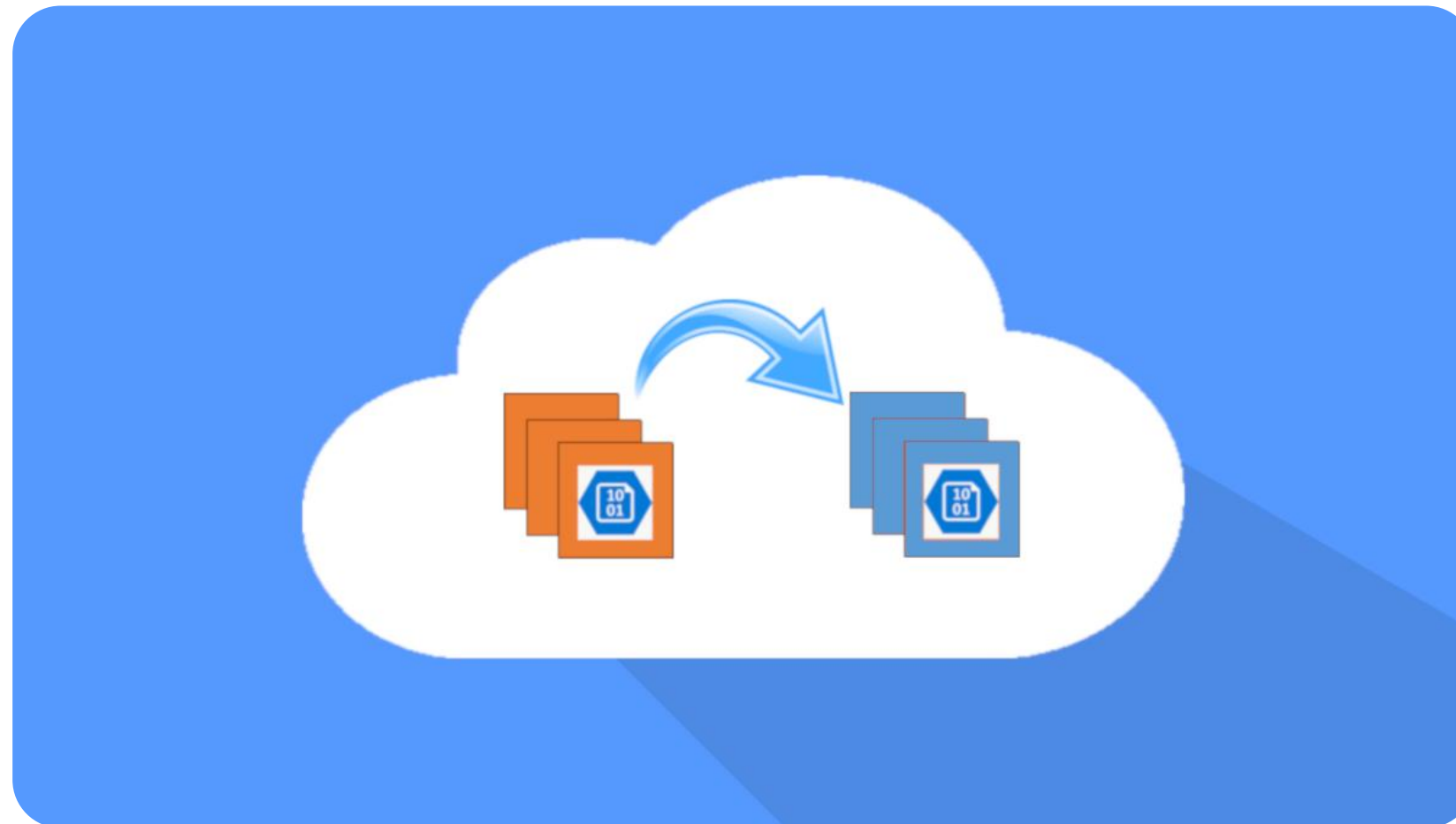
Storage Account Failover

Azure Storage supports account failover for geo-redundant storage accounts.

Type of failover	GRS/RA-GRS	GZRS/RA-GZRS
Customer-managed planned failover (preview)	General-purpose v2 accounts General-purpose v1 accounts Legacy Blob Storage accounts	General-purpose v2 accounts
Customer-managed (unplanned) failover	General-purpose v2 accounts General-purpose v1 accounts Legacy Blob Storage accounts	General-purpose v2 accounts
Microsoft-managed failover	All account types	General-purpose v2 accounts

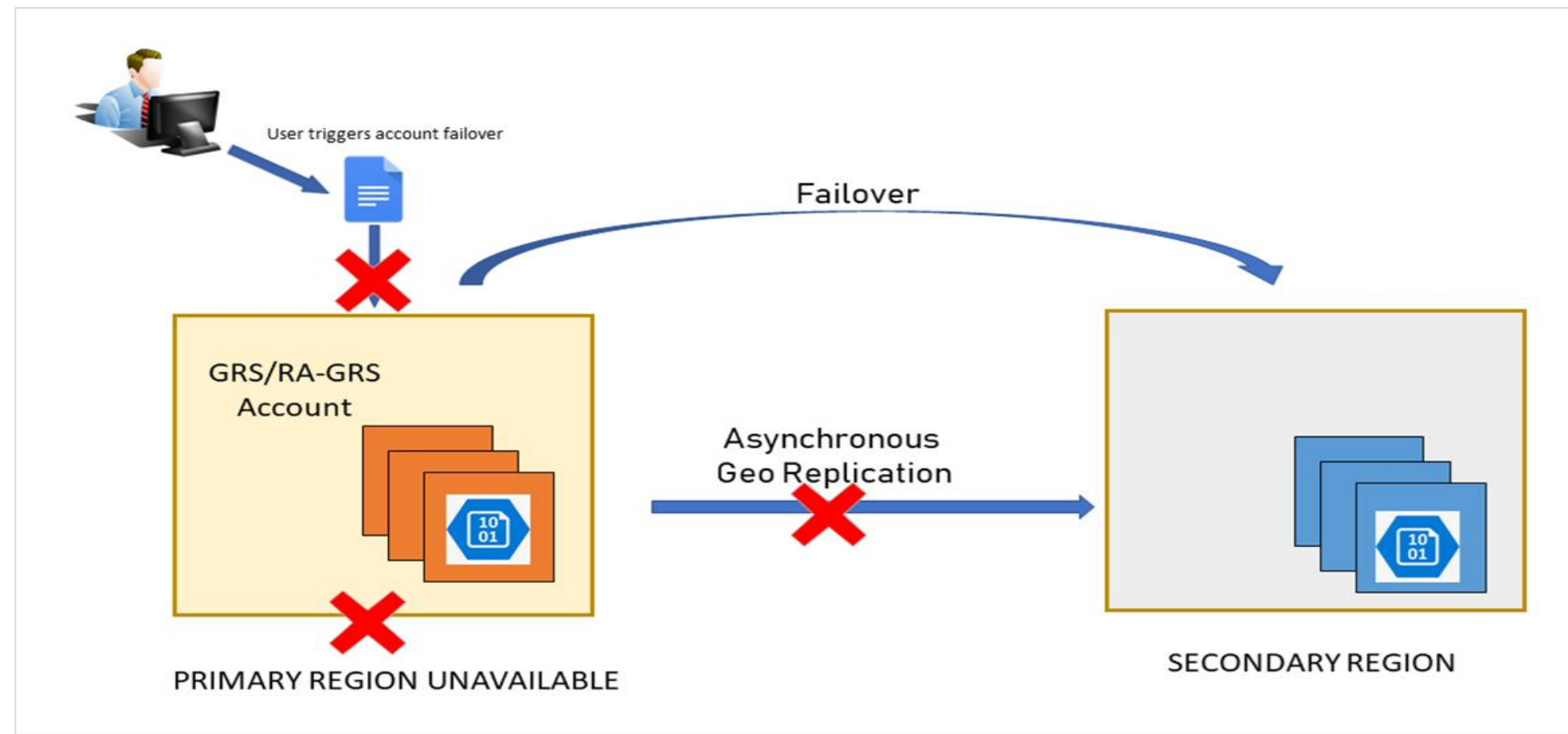
Storage Account Failover

If the primary endpoint becomes impossible to access, the user can start the storage account failover process.



Working of an Account Failover

The failover updates the secondary endpoint to become the primary endpoint for the storage account.



Once the failover is complete, clients can begin writing to the new primary endpoint.

Key Takeaways

- Availability Sets is a logical grouping of VMs within a data center.
- Azure Front Door is a global, scalable entry point that uses the Microsoft global edge network to create fast, secure, and highly scalable web applications.
- Storage account failover prevents unplanned service outages for the user.
- Azure virtual machine scale sets enable the user to deploy a set of identical VMs.
- The stateless storage layer serves as the distributed storage engine for database engine processes on compute replicas.





Knowledge Check

Knowledge Check

1

How many replication options are there while creating an Azure storage account?

- A. 2
- B. 5
- C. 8
- D. 4



Knowledge Check

1

How many replication options are there while creating an Azure storage account?

- A. 2
- B. 5
- C. 8
- D. 4



The correct answer is **D**

While creating the Azure Storage account, there are 4 replication options available: Locally redundant storage, Zone redundant storage, Geo-redundant storage, and Read-access Geo-redundant storage.

**Knowledge
Check**

2

Which scale is also known as scale-up and scale-down?

- A. Horizontal scale
- B. Vertical scale
- C. Inclined scale
- D. None of these



Knowledge
Check

2

Which scale is also known as scale-up and scale-down?

- A. Horizontal scale
- B. Vertical scale
- C. Inclined scale
- D. None of these



The correct answer is **B**

The Vertical scale is also known as scale-up and scale-down, which means increasing or decreasing virtual machine sizes in response to the workload.

**Knowledge
Check**

3

Which of the following account types is available for Storage Account Failover?

- A. General-purpose V1
- B. General-purpose V2
- C. Blob storage account
- D. All of these



**Knowledge
Check**

3

Which of the following account types is available for Storage Account Failover?

- A. General-purpose V1
- B. General-purpose V2
- C. Blob storage account
- D. All of these



The correct answer is **D**

Storage Account Failover is available for General-purpose V1, General-purpose V2, and Blob storage accounts.

**Knowledge
Check**

4

Which of the following defines scaling in and out?

- A. Horizontal scale
- B. Vertical scale
- C. Auto scale
- D. None of these



Knowledge
Check

4

Which of the following defines scaling in and out?

- A. Horizontal scale
- B. Vertical scale
- C. Auto scale
- D. None of these



The correct answer is **A**

Horizontal scaling defines scaling in and out.

**Knowledge
Check**

5

Which of the following availability models is based on the separation of compute and storage?

- A. High availability model
- B. Standard availability model
- C. Premium availability model
- D. None of these



**Knowledge
Check**

5

Which of the following availability models is based on the separation of compute and storage?

- A. High availability model
- B. Standard availability model
- C. Premium availability model
- D. None of these



The correct answer is **B**

The standard availability model is based on the separation of compute and storage.

Designing Solution for High Availability

Duration: 25 min.



Project Agenda: To design a solution for high availability for the given scenario

Description: You have been given a project to choose the right candidate service for a relational database that can be used for an OLTP application. You need to ensure that during an adverse event like a catastrophic failure or natural disaster, the users are still able to read the data. You need to configure Geo-replication so that you have a secondary database in place.

Perform the following:

1. Log in to the Azure account
2. Configure SQL Database and Geo-replication for a secondary database

TECHNOLOGY

Thank You